Chapter 42. Ventilator-Induced Lung Injury

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Ventilator-Induced Lung Injury: Introduction

The deleterious effects of mechanical ventilation on the lungs have long been referred to as *barotrauma.* For many years, clinicians defined barotrauma as the occurrence of air leaks resulting in the accumulation of extraalveolar air responsible for a number of manifestations, of which the most threatening is tension pneumothorax. In addition to these "macroscopic" events whose adverse consequences are usually immediately obvious, mechanical ventilation may produce more subtle physiologic and morphologic alterations, especially when it results in high airway pressures. Our knowledge of such alterations has stemmed mainly from experimental studies and has expanded considerably over recent years. Indeed, alterations in alveolar–capillary barrier integrity and release of both inflammatory and antiinflammatory mediators have been reported in animals ventilated with modalities resulting in high lung stretching. Tissue damage also may occur during mechanical ventilation when distal airways close and open repeatedly because of the movement of foam in the airway lumen or rupture of liquid menisci. The clinical relevance of these experimental findings received resounding confirmation with the results of the ARDS Network study, which showed a 22% reduction in mortality in patients with the acute respiratory distress syndrome through a simple reduction in tidal volume.[1]

The first comprehensive work demonstrating that mechanical ventilation may be unsafe in intact animals was performed by Webb and Tierney.[2] Rats ventilated with a peak inspiratory pressure (PIP) of 30 or 45 cm H_2O displayed pulmonary edema within 20 minutes to 1 hour, depending on the pressure level. Microscopic examination of the lungs disclosed moderate interstitial edema in animals ventilated with the lower peak pressure, contrasting with profuse edema and alveolar flooding in animals ventilated with the highest PIP. Other studies subsequently documented the occurrence of pulmonary edema and lung ultrastructural abnormalities[3,4] after even very short periods of intermittent positive-inspiratory pressure with high PIP. Kolobow et al[5] reported progressive lung injury in sheep ventilated with 50 cm H_2O PIP over 48 hours, manifested as decreased pulmonary compliance and deterioration in blood oxygenation. Some of the animals died before the 48-hour end point. At autopsy, lungs exhibited congestion and severe atelectasis.

The mechanisms underlying this ventilator-induced lung injury (VILI) have been for the most part elucidated. Two main factors explain its development: the magnitude of lung overdistension and its duration. Ventilation with very high peak transalveolar pressure results in acute, rapidly fatal, permeability pulmonary edema, whereas more protracted ventilation involving alveolar distension of lesser magnitude produces a lung injury in which inflammatory phenomena may play a role. This chapter presents the current knowledge on VILI. A Medline research employing "ventilator-induced lung injury" as keywords and limited to English articles yielded approximately 1900 results up to January 2005 (when the chapter in the second edition of this book was written). A Medline search employing the same keywords yielded 3860 results when updated to December 2010, reflecting the huge scientific productivity in this field over the past 6 years. To provide the most recent findings to readers, we have condensed material that was presented in greater detail in the second edition of this book.

Mechanisms of Ventilator-Induced Lung Injury

Ventilation modalities with very high distending pressures (typically >30 cm H_2O, depending on the species) result in pulmonary edema. Both increased filtration and alteration in capillary permeability participate in edema formation. The reasons for these abnormalities have been partly elucidated.

Evidence in Support of Increased Filtration

The first hypothesis put forward to account for ventilator-induced edema involved hydrostatic alterations.[2,6] Parker et al[6] calculated that mean lung microvascular pressure increased by 12.5 cm H_2O during ventilation of open-chest dogs at 64 cm H_2O PIP. Increased filtration may result from an increase of pulmonary capillary pressure, a decrease in lung interstitial pressure, or both. Surfactant inactivation and increased lung volume participate in the decrease in lung interstitial pressure.

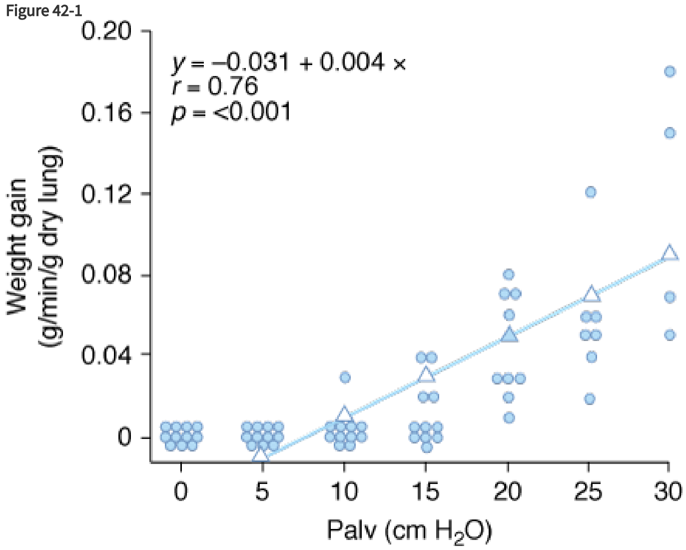
The increase in alveolar surface tension resulting from surfactant inactivation would be expected to further decrease the negative pressures surrounding alveolar vessels, thereby increasing vascular transmural pressure and enhancing fluid filtration. Faridy et al[7] showed that ventilating excised dog lungs altered pulmonary pressure–volume curves and increased surface tension of lung extracts commensurately with the magnitude of tidal volume and duration of ventilation. Comparable findings were reported in excised rat lungs.[8,9] These anomalies were ascribed to depletion or inactivation of surfactant. Of note, surfactant alterations failed to occur when positive end-expiratory pressure (PEEP) was used.[7,8,10,11]

Pattle[12] and Clements[13] suggested that an increase in alveolar surface tension might increase filtration. Albert et al[14] found that the isogravimetric pressure (the vascular pressure at which net fluid flux from pulmonary vessels is zero) decreased when alveolar surface tension was increased by cooling and ventilating lungs at a low resting volume in open chest dogs. The authors interpreted this fall in isogravimetric pressure as indicative of a fall in perimicrovascular pressure.

Aerosolized ^{99m}Tc-DTPA (technetium-99m diethylenetriamine pentaacetic acid) clearance was found to increase following detergent aerosolization in rabbits[15] and dogs.[16] This finding was ascribed to regional overexpansion secondary to uneven lung inflation during mechanical ventilation of alveoli with altered surface tension rather than to elimination of peculiar barrier properties of surfactant.[16]

Increased filtration during mechanical ventilation probably also occurs at the extraalveolar level. During lung inflation, because of "pulmonary interdependence," the pressure in the perivascular space surrounding extraalveolar vessels decreases, which, in turn, increases transmural pressure. This effect of lung volume on pulmonary vessels is well documented.[17] Inflating the lungs dilates the extraalveolar vessels.[18] During inflation from a low transpulmonary pressure, the increase in vessel diameter is such that an effective outward-acting pressure in excess of pleural pressure (1 to 2 cm H_2O for each centimeter of water increase in transpulmonary pressure) expands the vessels.[19] The potential importance of fluid leakage through extraalveolar vessels has been established in both excised lungs[20] and in situ lungs of open-chest animals.[21] Moreover, inflation of in situ lobes under zone 1 conditions (i.e., where alveolar pressure exceeds pulmonary arterial and venous pressures) was found to precipitate hydrostatic pulmonary edema in dogs.[22] Because there is no flow in capillaries under zone 1 conditions, it is likely that the edema fluid leaked from distended extraalveolar vessels. The rate of edema formation was significantly correlated with the level of alveolar pressure and, therefore, the magnitude of distension (Fig. 42-1).

Figure 42-1



Relationship between alveolar pressure (Palv) and rate of hydrostatic edema formation in in situ lobes of open-chest dogs. Pulmonary arterial and venous pressures were kept at 1 cm H_2O. When lung volume was low (Palv = 10 cm H_2O), no edema occurred, whereas greater distensions produced a linear increase in lung weight gain, indicating edema formation. (Used, with permission, from Albert et al.[22])

The effects of decreased perimicrovascular pressure and surfactant alterations probably combine with increased intravascular pressure to augment transmural microvascular pressure during overinflation. Although the magnitude of these changes is difficult to evaluate with precision, the increase in vascular pressure seems relatively modest and is unlikely to explain per se the rapid development of profuse edema (especially in small animals) observed after high PIP ventilation.

Evidence in Support of Permeability Alterations

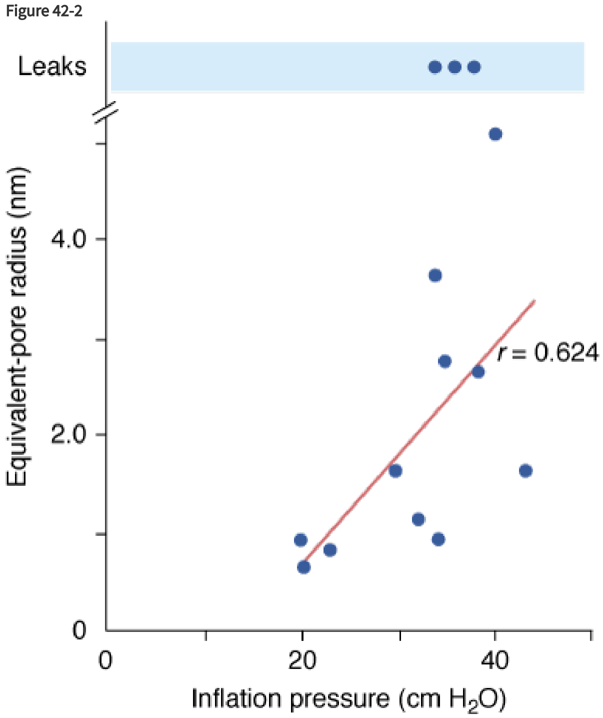
Many experimental studies in isolated lungs, as well as in open-chest or intact animals, have demonstrated permeability alterations (in response to high airway pressures) in the epithelium and, more unexpectedly, the endothelium.

Epithelial Permeability Changes in Response to High Airway Pressure

The increase in alveolar epithelial permeability to small hydrophilic solutes observed when lung volume increases is a physiologic phenomenon. Elevation of functional residual capacity (FRC), in sheep, obtained by increasing the level of PEEP during mechanical ventilation[23] or spontaneous ventilation,[24] was associated with an increase in aerosolized diethylenetriamine pentaacetic acid (DTPA) clearance. Clearance augmentation was larger than expected from the changes in alveolar exchange surface area.

Effects of overinflation on epithelial permeability were studied extensively by Egan during static inflation of fluid filled in situ lobes.[25,26] The equivalent-pore approach was used to describe the permeability of the epithelium to hydrophilic solutes of various sizes. Equivalent-pore radii increased from about 1 nm at 20 cm H_2O inflating pressure to 5 nm at 40 cm H_2O alveolar pressure. In some instances, free diffusion of albumin across the epithelium was observed, indicating the presence of large leaks (Fig. 42-2). Permeability alterations persisted or even increased after cessation of inflation, suggesting irreversible epithelial injury. High airway pressures applied to in situ lobes, however, resulted in supraphysiologic overinflation because of the compression of adjacent parenchyma. Egan[27] performed experiments under less extreme conditions in rabbits, in which both in situ lobes and entire lungs were distended with 40 cm H_2O airway pressure. Static segmental inflation resulted in a sixfold to 12-fold increase in lung volume from FRC and in an epithelium permeable to solutes such as albumin, cytochrome c, and cyanocobalamin. In contrast, whole-lung inflation resulted in only a threefold to fourfold increase in lung volume and a lesser rate of escape of the smaller solutes from the alveolar spaces. The increase in albumin permeability resulting from whole-lung overinflation was negligible. Hence, only major increases in lung volume produce significant changes in permeability to large molecules.

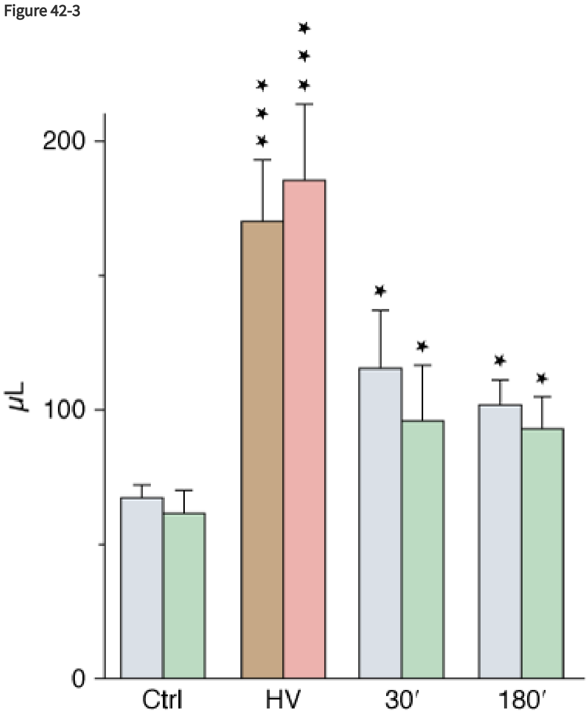
Figure 42-2



Effect of inflation pressure on the epithelial permeability of fluid-filled in situ lobes of sheep lung. Permeability is characterized by an equivalent-pore radius. A linear relationship was observed between inflation pressure and pore radius. At the highest levels of inflation, free diffusion of albumin was sometimes observed, indicating the presence of large leaks. (Used, with permission, from Egan et al.[25])

Although less-well-documented on a quantitative basis, epithelial alveolar permeability alterations are probably present in varying degrees during ventilator-induced pulmonary edema. During positive pressure-ventilation with 41 cm H_2O PIP, increases in alveolar permeability to small (DTPA), but not large (albumin), solutes were observed after 8 hours by Ramanathan et al in lambs.[28] After 2 minutes of high-pressure ventilation, epithelial lining fluid (ELF) volume calculated from bronchoalveolar lavage increased by 180%[4] (Fig. 42-3). ELF protein concentration decreased (Fig. 42-4*A*) even though protein content increased by 76% (Fig. 42-4*B*), suggesting that most of the excess fluid entered the alveolar-airway lumen through increased convection. In contrast to the severely altered sieving properties of microvessels after only 2 minutes of overinflation, epithelial permeability appears to be better preserved. This speculation is consistent with the scarcity of epithelial cell alterations on electron microscopic examination, contrasting with the widespread endothelial abnormalities.[4] The presence of blood cells in the alveoli, however, suggested a few large endothelial and epithelial tears. During longer (2 hours) ventilation periods with a lower tidal volume (V_T) (19 mL/kg in rats, resulting in a PIP of 19 cm H_2O), the presence of secretory Clara-cell protein (a protein found in airway lumen) in the vasculature contrasting with a decrease in its concentration in bronchoalveolar lavage fluid further suggests an increase in alveolar barrier permeability.[29] Secretory Clara-cell protein was also found in the plasma of mice ventilated with 35 cm H_2O PIP for 2 hours and even earlier (30 minutes) when PIP was 55 cm H_2O.[30] Resorption of alveolar liquid by distal airway epithelium is decreased during VILI because of depressed sodium transport mechanisms.[31] It can be restored by β-adrenergic stimulation that recruits ion-transporting proteins to the plasma membrane of alveolar epithelial cells or by β_1-sodium-potassium adenosine triphosphatase (ATPase) subunit gene transfer.[32,33]

Figure 42-3



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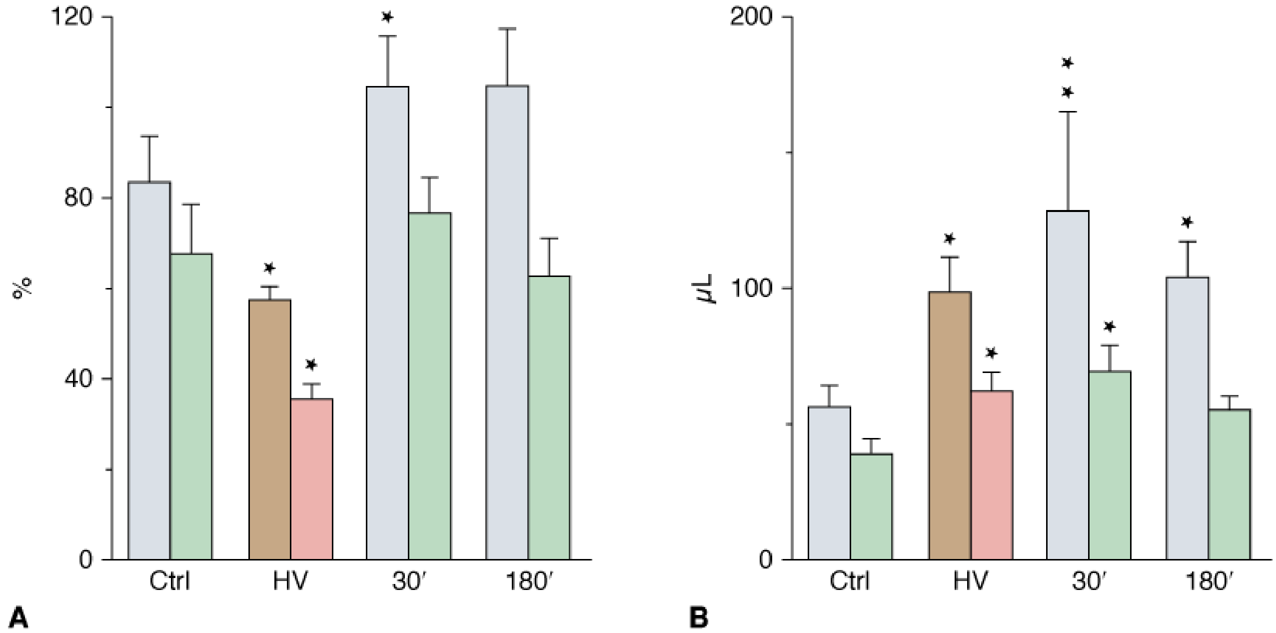
Estimates of epithelial lining fluid (ELF) volume by bronchoalveolar lavage after ventilation with 45 cm H_2O peak airway pressure for 2 minutes (*HV*) and recovery. Two successive lavages were performed (*grey and brown bars*, first lavage; *green and pink bars*, second lavage). ELF volume increased significantly after HV and then decreased during recovery but remained higher than in control subjects (*Ctrl*). * = $p < 0.05$; *** = $p < 0.001$, as compared to controls. (Used, with permission, from Dreyfuss et al.[4])

Figure 42-4

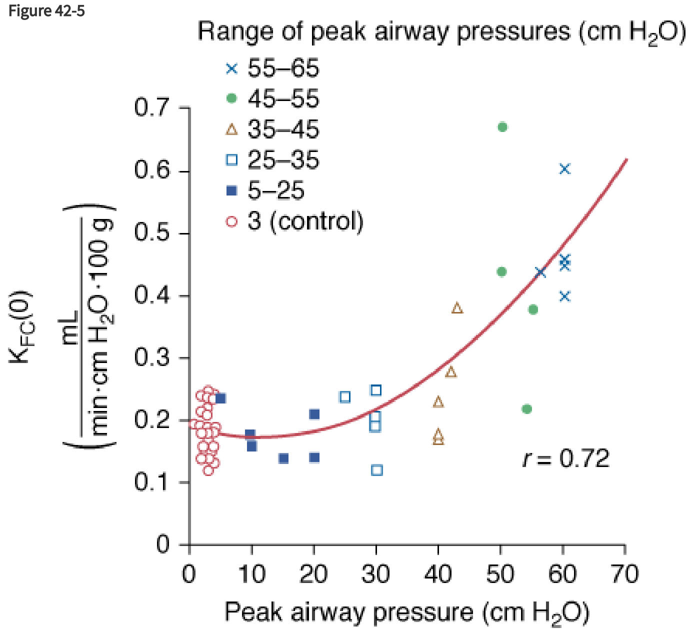


A. Protein concentration and (**B**) protein content in epithelial lining fluid (see Fig. 42-3 for details on the lavages). Protein content (expressed as the equivalent plasma volume) increased after high-pressure ventilation (*HV*) and then remained unchanged during recovery. Protein concentration (as a percentage of plasma protein concentration) decreased after HV and returned to normal during recovery (see text for details). * = $p < 0.05$; ** = $p < 0.01$, as compared to controls. (Used, with permission, from Dreyfuss et al.[4])

Microvascular Permeability Alterations during Mechanical Ventilation-Induced Pulmonary Edema

In a study in isolated blood-perfused lobes from dogs, Parker et al[34] demonstrated that ventilation for 20 minutes with graded increases in PIP did not affect microvascular permeability up to 30 cm H_2O. Higher PIP (45 to 65 cm H_2O) was associated with increases in the capillary filtration coefficient (Fig. 42-5) and decreases in isogravimetric capillary pressure, suggesting the existence of an airway pressure threshold. Similarly, the estimated protein reflection coefficient was decreased only at the highest airway pressures. Of note, at an airway pressure above the threshold, the increase in capillary filtration coefficient occurred immediately and continued in some lobes after cessation of distension.

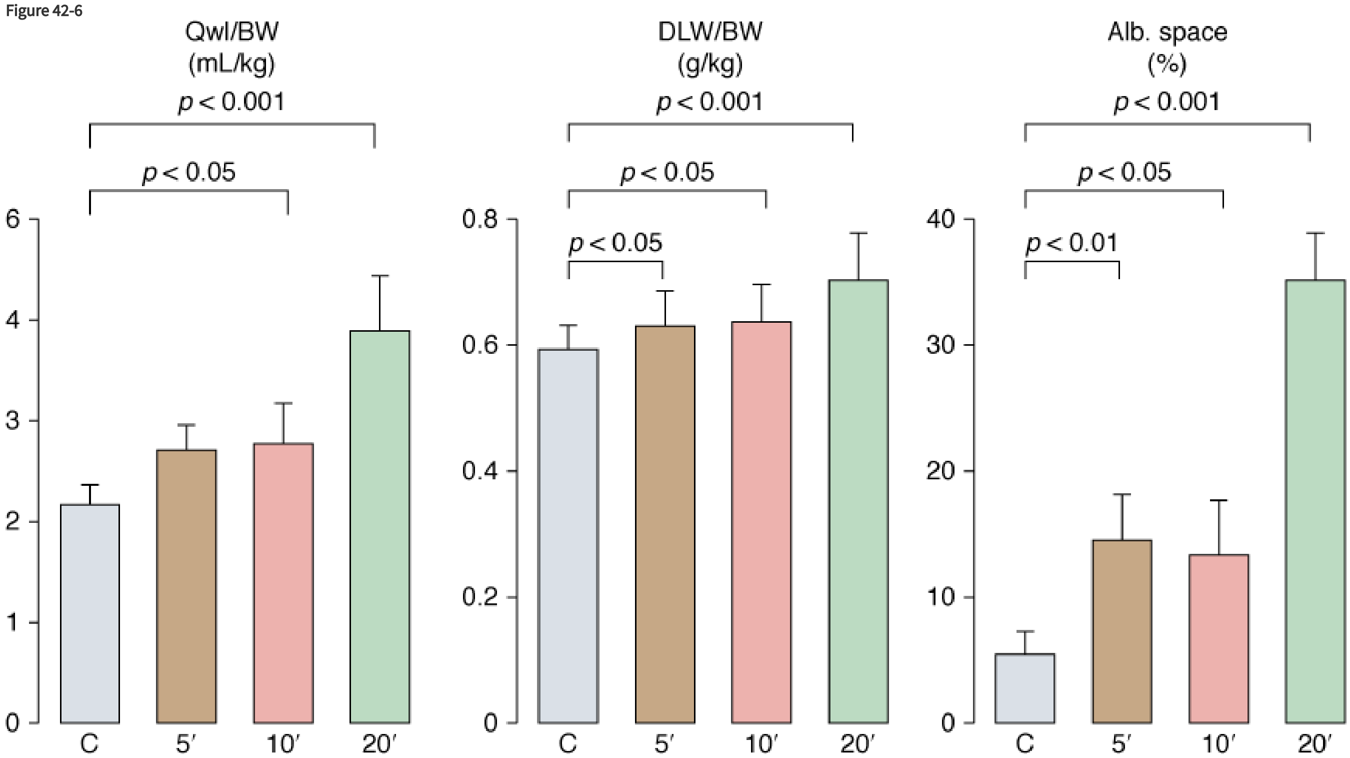
Figure 42-5



Effect of peak airway pressure on the capillary filtration coefficient (K_{FC}) of isolated, blood-perfused lobes of dog lungs receiving 20 minutes of intermittent positive-pressure ventilation. Moderate (up to 30 cm H_2O) increases in peak pressure did not affect K_{FC}, whereas higher levels of peak pressure resulted in a very steep increase in K_{FC}. (Used, with permission, from Parker et al.[34])

To evaluate permeability alterations in intact animals, extravascular lung water and bloodless dry-lung weight and the distribution space in lungs of ^{125}I-labeled albumin injected in the systemic circulation were measured in intact rats subjected to 45 cm H_2O PIP ventilation.[3] Pulmonary edema developed very rapidly and was easily demonstrable after only 5 to 10 minutes of high-pressure ventilation (Fig. 42-6). Light microscopic examination revealed that, at this stage, edema fluid remained confined to interstitial spaces, where it accumulated in the large peribronchovascular cuffs. No alveolar flooding was apparent. The presence of permeability alterations was indicated by a significant increase in dry-lung weight and of albumin distribution space (Fig. 42-6). After 10 minutes of high PIP ventilation, pulmonary edema was more abundant but still involved only the interstitial spaces. After 20 minutes of ventilation, findings were strikingly different, with tracheal flooding in all animals. Widespread alveolar flooding was obvious upon light microscopic examination. The severity of the permeability alterations was such that the ratio of ^{125}I-albumin activity in tracheal fluid versus plasma was close to unity, indicating the loss of permselectivity of the capillary barrier. The severity of the permeability defect was indicated by the relationship between dry-lung weight and extravascular lung water (Fig. 42-7), which indicated that the concentration of protein in extravasated fluid was high, suggesting the presence of numerous large capillary leaks. Similar findings relating the duration of high-PIP ventilation and alterations in capillary permeability have been made in mice.[30] This increase in endothelium permeability occurs both at the alveolar and extraalveolar level.[35]

Figure 42-6



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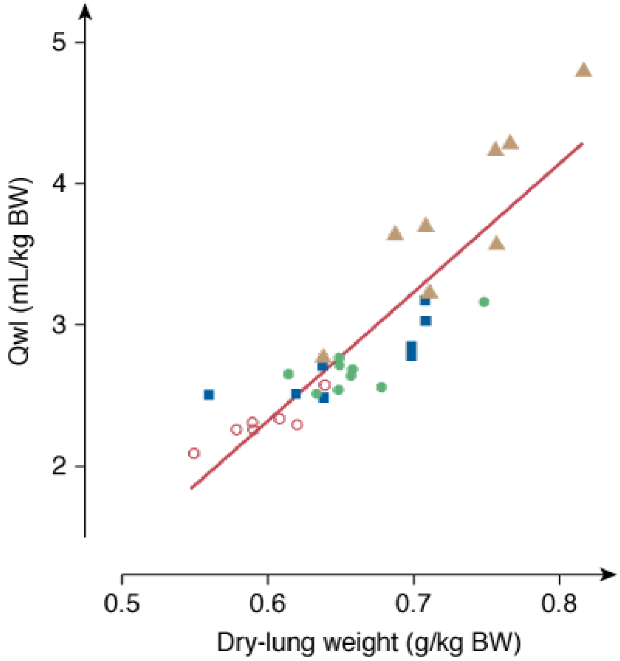
Effect of 45-cm H_2O peak airway pressure ventilation in intact rats. Pulmonary edema was assessed by the determination of extravascular lung water content (*Qwl/BW*) and permeability alterations by the determination of bloodless dry lung weight (*DLW/BW*) and of the distribution space in the lungs of ^{125}I-labeled albumin (*Alb. Space*). Permeability pulmonary edema developed rapidly (5 minutes). After 20 minutes of mechanical ventilation, there was a dramatic increase in all indexes ($p < 0.01$ versus other groups). (Adapted, with permission, from Dreyfuss et al.[3])

Figure 42-7



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Relationship between extravascular lung water content (*Qwl/BW*) and dry lung weight during mechanical ventilation with peak airway pressure 45 cm H_2O. The concentration of proteins in the edema fluid estimated from this relationship was consistent with the outpouring of protein-rich edema fluid reflecting severe permeability alterations. *Open circles,* control conditions; *closed circles,* 5 minutes of ventilation; *closed squares,* 10 minutes of ventilation; *closed triangles,* 20 minutes of ventilation. (Used, with permission, from Dreyfuss et al.[3])

The time course of edema development depends on the size of the species: in rats, high PIP ventilation for as little as 2 minutes is sufficient to produce permeability edema,[4] although alterations are minor and result in fairly mild edema. The increase in the ratio of extravascular lung water to blood-free dry-lung weight was increased by only 17% after 2 minutes[4] versus 90% when the duration of the challenge was increased up to 20 minutes.[36] In larger animals, considerably longer durations of high PIP ventilation are required to produce significant alterations. For instance, in lambs mechanically ventilated at 58 cm H_2O PIP for 6 hours, Carlton et al[37] found an increase in the ratio of extravascular lung water to dry-lung weight of only 19%. Results of histologic lung studies were either normal or showed only mild perivascular edema with no alveolar edema. In a study by Borelli et al,[38] sheep mechanically ventilated for 18 hours with a PIP of 50 cm H_2O developed both interstitial and alveolar edema.

Wet-lung weight normalized for body weight increased by 89% compared to normal lungs.[39] After 27 hours of ventilation, pathologic abnormalities were much more marked, and lung weight was increased by 136% compared to normal lungs. Clearly, the ventilation time needed to produce severe edema is less than 1 hour in small species but more than 24 hours in large animals. Blood-gas barrier thickness is an important component of capillary resistance to mechanical stretch, and is more important in larger animals.[40] Thus, one reason why permeability alterations were not consistently found is that they may become detectable only after longer ventilation times, in contrast to the immediate occurrence of microvascular pressure anomalies.

In summary, increased microvascular filtration pressure and altered microvascular permeability probably compound their effects to produce high PIP pulmonary edema. Although the hydrostatic component seems to be moderate on a quantitative basis, at least in closed-chest animals, it may nevertheless have a substantial impact. Indeed, in the face of a microvascular barrier with altered sieving properties, any increase in driving pressure will have a dramatic effect on edema formation.[41–43]

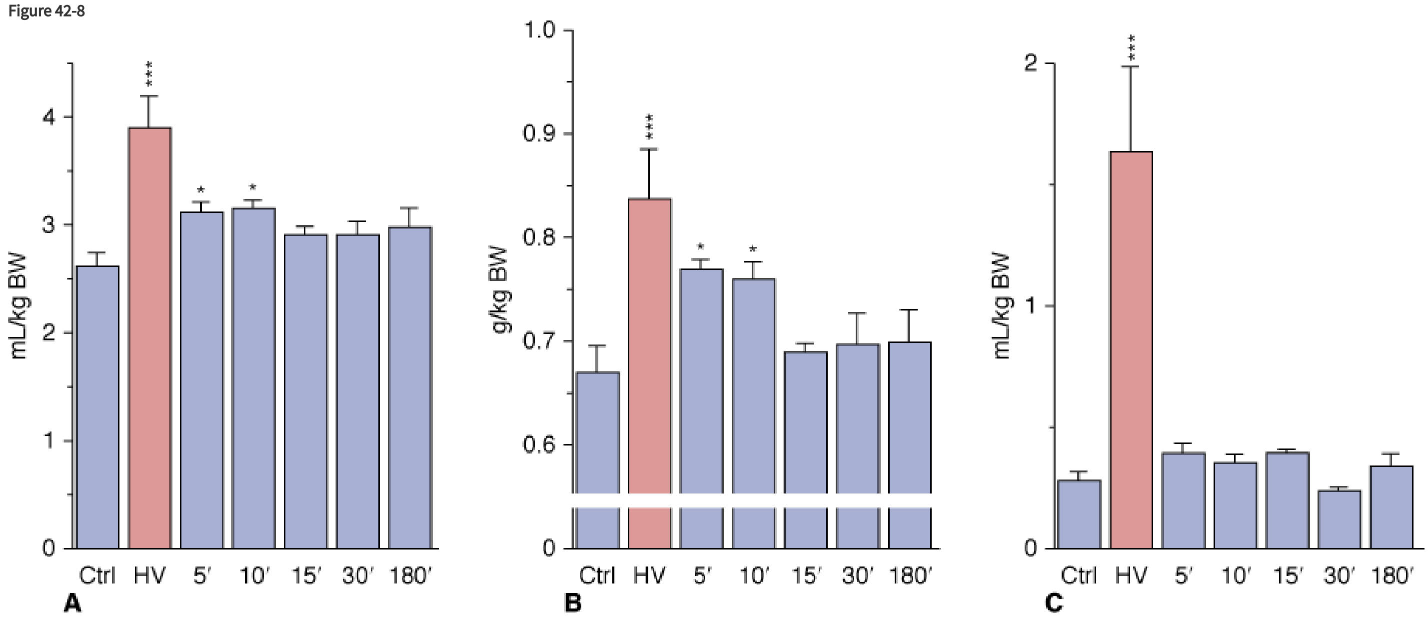
Mechanisms of Acute Increase in Alveolo-Capillary Permeability during Lung Overinflation

Interest has increasingly focused on the cellular response to mechanical strain and this subject has been comprehensively reviewed.[44] Tschumperlin and Margulies[45] observed increased cell death when alveolar epithelial cells were submitted in vitro to deformation. Cyclic deformation led to significantly greater cell death than did static deformation. The percentage of dead cells after 1 day ranged from 0.5% to

72% depending on the changes in surface area (up to 50%). In cyclically deformed cells, injury occurred rapidly, with most of cell death occurring during the first 5 minutes of deformation.[46] These authors showed that basement membrane surface area increased approximately 40% when lung volume was varied from FRC to total lung capacity.[47] These increases suggest that epithelial cells undergo significant stretch during large inflations. Vlahakis et al[48,49] labeled membrane lipids to study deformation-induced lipid trafficking and studied in a direct manner (laser confocal microscopy) the response of epithelial cells of the alveolar basement membrane response to forces. A 25% stretch deformation resulted in lipid transport to the plasma membrane that allowed maintenance of its integrity and an increase in epithelial cell surface area. Such lipid trafficking occurred in all cells, whereas plasma breaks were seen in only a small percentage of cells. The authors concluded that deformation-induced lipid trafficking serves, in part, to repair plasma breaks so as to maintain plasma membrane integrity and cell viability, and that this could be viewed as a cytoprotective mechanism against the plasma membrane stress failure seen during VILI.[36,50]

Overinflating the lungs results in the increase in epithelial[25] and endothelial[3,34] permeability, both occurring after inflating the lung above the same end-inspiratory pressure threshold.[51] Using an isolated perfused rat model, Parker et al showed that gadolinium (which blocks stretch-activated nonselective cation channels) annulled the increase in microvascular permeability induced by high PIP.[52] This suggests that entry through stretch-activated channels and an increase in intracellular calcium ion (Ca^{2+}) concentration might initiate the increase in permeability. This increase results in the activation of tyrosine kinases,[53] activation of the Ca^{2+}–calmodulin pathway and phosphorylation of myosin light chain kinase.[54] Taken together, these results suggest that the increase in microvascular permeability below the cell rupture point, which occurs under extreme stretch conditions ("stress failure"[50,55]), is not simply a passive physical phenomenon, but the result of biochemical reactions. In vitro studies show that cell plasticity and deformation-induced lipid trafficking protect against strain injury. Interventions that impair deformation-induced lipid trafficking reduce the likelihood of plasma membrane resealing[49,56] and increase cell death. It may help explain why the breaks seen across endothelial cells, after short periods of high transmural capillary pressure[57] or overinflation[4] are transient. In this latter study, rats were subjected to mechanical ventilation with 35 mm Hg PIP for only 2 minutes and allowed to recover for graded periods of up to 3 hours. The animals that were killed immediately after the challenge exhibited mild pulmonary edema with severe permeability alterations, as attested by increases in dry-lung weight and albumin space (Fig. 42-8). The very rapid occurrence of microvascular injury was ascertained by the ultrastructural study, which disclosed changes identical with those previously described after longer periods of ventilation.

Figure 42-8



Effect of 45 cm H_2O peak airway pressure ventilation for 2 minutes (*HV*) and followed by recovery for various lengths of time in intact rats. **A.** Extravascular lung water. **B.** Dry lung weight. **C.** Albumin distribution space. Permeability edema already present after 2 minutes cleared rapidly after cessation of ventilation. * = $p < 0.05$; *** = $p < 0.001$, as compared to controls. (Used, with permission, from Dreyfuss et al.[4])

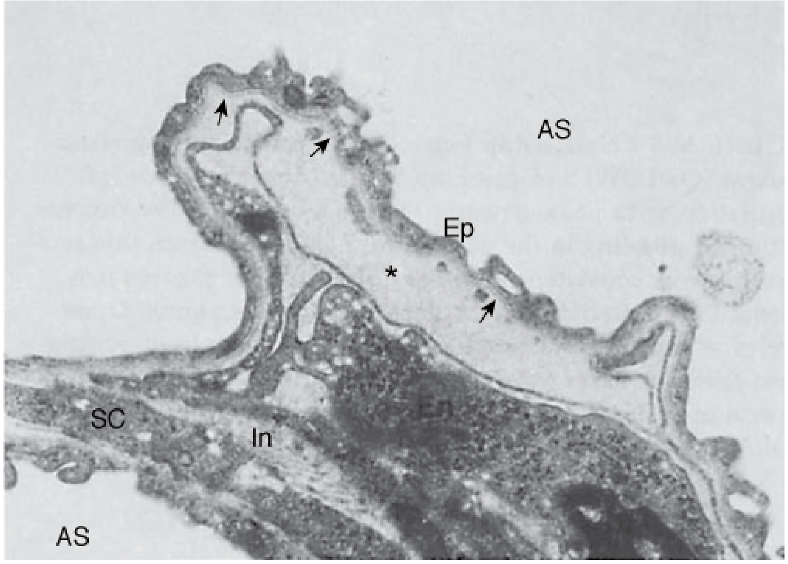
These data strongly suggest that, at least in small animals, vascular leakage after overinflation is almost immediate and extensive. During recovery, both extravascular lung water and dry lung weight promptly returned to normal (see Fig. 42-8A and B), indicating that resorption of edema can be very rapid. Edema resorption does not necessarily indicate restoration of the alveolo–capillary barrier. The 30-minute distribution space of ^{125}I-albumin in lungs, however, was found to be normal, indicating that no albumin permeability defect was present after the time of tracer injection (see Fig. 42-8C). Because the tracer was injected during the recovery period (i.e., after cessation of overinflation), this finding demonstrated reversal of the alterations in permeability. ELF volume decreased after cessation of overinflation (see Fig. 42-3), reflecting resorption of the excess alveolar fluid concomitantly with the decrease in extravascular lung water. Even after 3 hours of recovery, however, ELF volume remained higher than in control animals. No marked changes in epithelial fluid protein content were observed (see Fig. 42-4B), reflecting the slower-than-water clearance of protein in alveolar edema fluid.[58] The absence of notable protein resorption explains why the protein concentration in ELF increased during recovery (see Fig. 42-4A).

Ultrastructural Findings of Ventilator-Induced Lung Injury

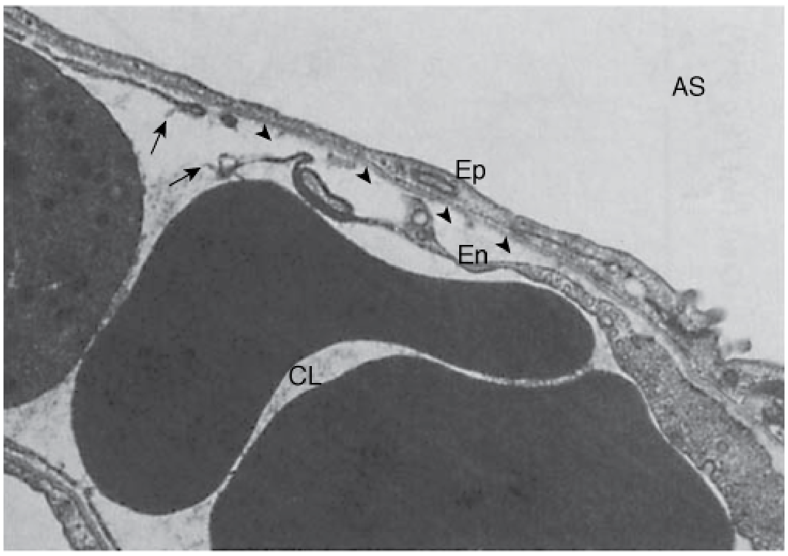
Electron microscopic studies have confirmed that permeability alterations are prominent in the genesis of ventilator-associated pulmonary edema. Both endothelial and epithelial alterations have been observed.[3,4,30,36]

After short durations (5 to 10 minutes) of 45 cm H_2O PIP mechanical ventilation in rats, striking capillary abnormalities consistent with pulmonary edema of the nonhydrostatic type were observed. Some endothelial cells were detached from their basement membrane, resulting in the formation of intracapillary blebs filled with plasma-like material (Fig. 42-9A). Endothelial cells exhibited focal disruptions (Fig. 42-9B). Bleb formation has been reported in experimental high-permeability edema, regardless of the nature of the causative agent, and in acute respiratory distress syndrome (ARDS),[59–62] but not in experimental hydrostatic pulmonary edema.[59,60,63] Longer durations (20 minutes) of high PIP ventilation in rats resulted in alveolar flooding and obvious permeability alterations. Pathologic studies showed that this severe edema was accompanied with diffuse damage to the alveolar–capillary barrier. In addition to the capillary lesions, ultrastructural studies disclosed profound alterations in the epithelial layer. The severity of alterations varied. In some sites, the epithelial lining appeared intact. In many areas, however, findings included discontinuities (Fig. 42-10A and B) and sometimes almost complete destruction of type I cells, leaving a denuded basement membrane (Fig. 42-10B and C). In contrast, type II cells appeared preserved. Hyaline membranes filled the alveolar spaces in many of the sections examined (Fig. 42-10C). Similar to the endothelial abnormalities, these lesions are not specific and occur in toxic injuries as well as in ARDS.[59–62] In contrast, when pressures remain in the 30- to 40-mm Hg range, hydrostatic edema does not affect epithelial integrity.[59,60,64]

Figure 42-9



A



B

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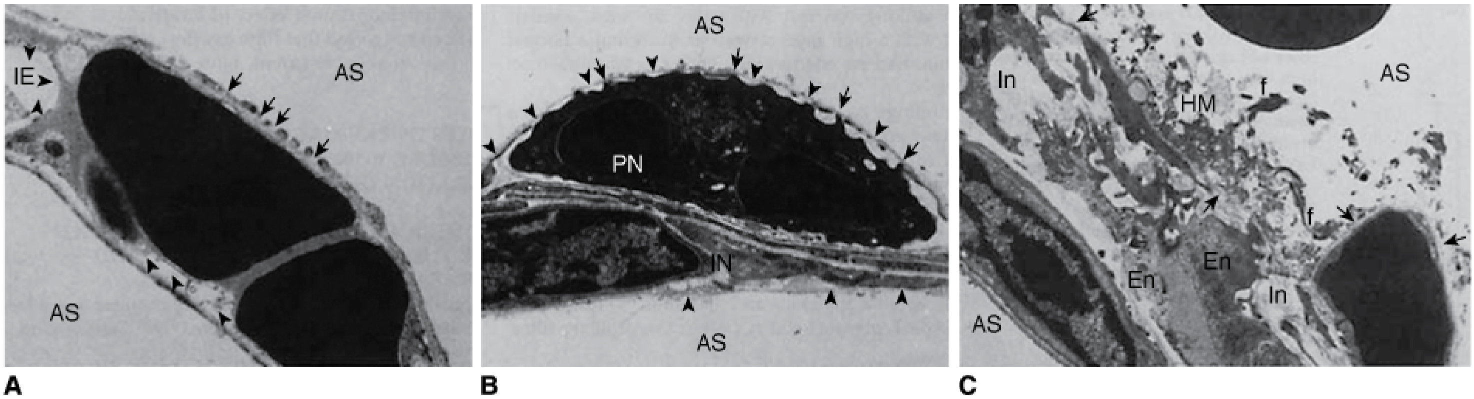
Ultrastructural features of the blood–air barrier after 5 minutes of 45 cm H_2O peak airway pressure in a closed chest rat. **A.** The most striking change is the formation of an endothelial bleb secondary to detachment of the thin part of the endothelial cell (*En*) from the basement membrane (*arrows*). This bleb is filled with electron-dense material (*) of the same density as plasma. At this stage, epithelial type I cells (*Ep*) are intact. Note interstitial edema (*In*). *AS,* alveolar space; *SC,* septal cell. (Used, with permission, from Dreyfuss et al.[3]) **B.** *Arrows* indicate a disruption of the thin part of the endothelial cell detached from the basement membrane (*arrowheads*). *CL,* capillary lumen. (Courtesy Paul Soler, Ph.D.)

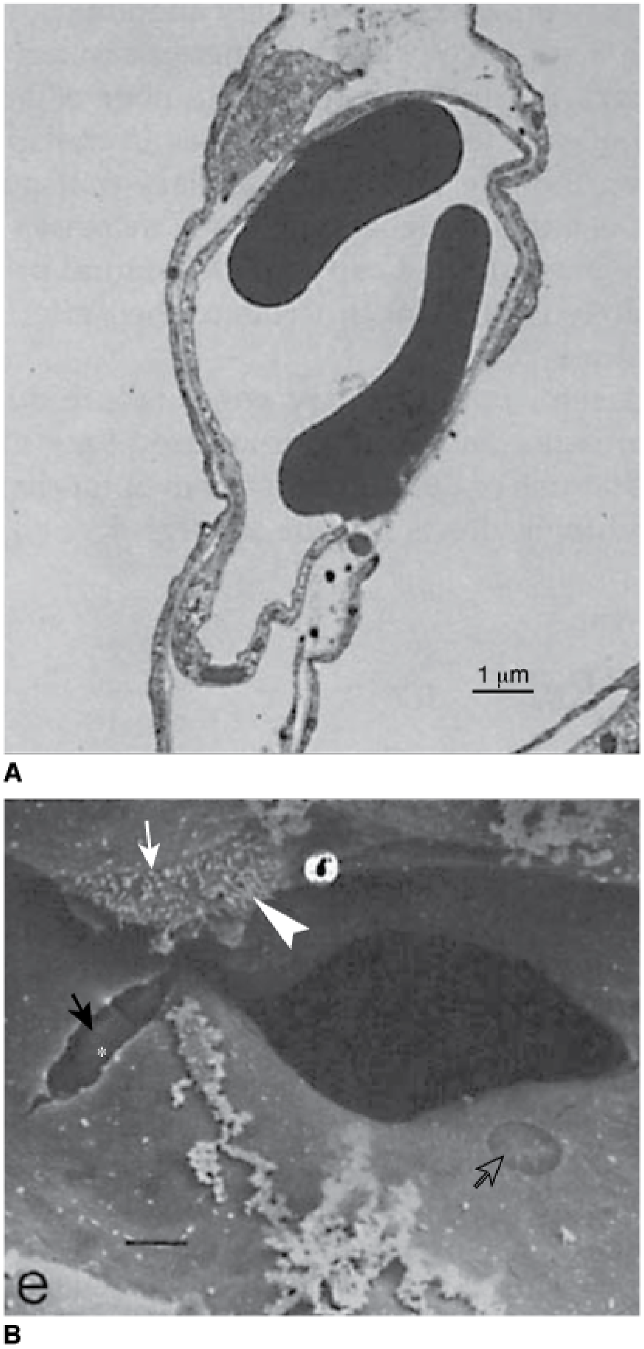
Figure 42-10



Ultrastructural features of the blood–air barrier after 20 minutes of 45 cm H_2O peak airway pressure in a closed chest rat. **A.** Type I cells show numerous gaps (*arrows*). *Arrowheads* point at the basement membrane from which the endothelial cell is detached. *AS,* alveolar space; *IE,* interstitial edema. (Courtesy Dr. Paul Soler.) **B.** Epithelial and endothelial cells are almost completely destroyed, resulting in denuded basement membranes (*arrowheads*). A polymorphonuclear neutrophil (*PN*) inside the capillary lumen exhibits cytoplasmic processes, protruding through gaps in the capillary endothelium. *IN,* interstitium. (Courtesy Dr. Paul Soler.) **C.** Very severe alteration of the alveolar capillary barrier. Complete lysis of the epithelial layer (*upper right quadrant*) results in denudation of the basement membrane (*arrows*). Alveolar space is occupied by hyaline membranes (*HM*) composed of cell debris and fibrin (*f*). *En,* endothelial cells; *In,* interstitial edema. (Used, with permission, from Dreyfuss et al.[3])

Studies using both transmission and scanning electron microscopy have demonstrated ultrastructural breaks in endothelial and epithelial cells when capillary transmural pressure was raised to 40 mm Hg or more.[55,65,66] These breaks are the morphologic correlates of the increased microvascular permeability reported with very high microvascular pressures[67,68] and described as the stretched-pore phenomenon.[69,70] The similarities between capillary stress failure and VILI are quite remarkable. First, the electron microscopic appearance of endothelial and epithelial cell lesions exhibit similarities (Fig. 42-11).[55,65] Second, both types of damage are partly reversible. Albumin leakage from capillaries during high-volume ventilation ceased almost immediately after discontinuation of ventilation.[4] When capillary pressures were lowered to normal after elevation to a level causing stress failure, the number of endothelial and epithelial breaks fell compared with control experiments in which pressure remained elevated.[57] More importantly, capillary stress failure is influenced by lung inflation: At a capillary transmural pressure of 32.5 cm H_2O, increasing lung volume from a transpulmonary pressure of 5 to 20 cm H_2O resulted in a significant increase in the number of capillary endothelium and alveolar epithelium breaks (Fig. 42-12).[50] Thus, vascular pressures that are too low to affect microvascular permeability when lung volume is normal may produce permeability alterations when combined with a sufficiently marked increase in lung volume.

Figure 42-11



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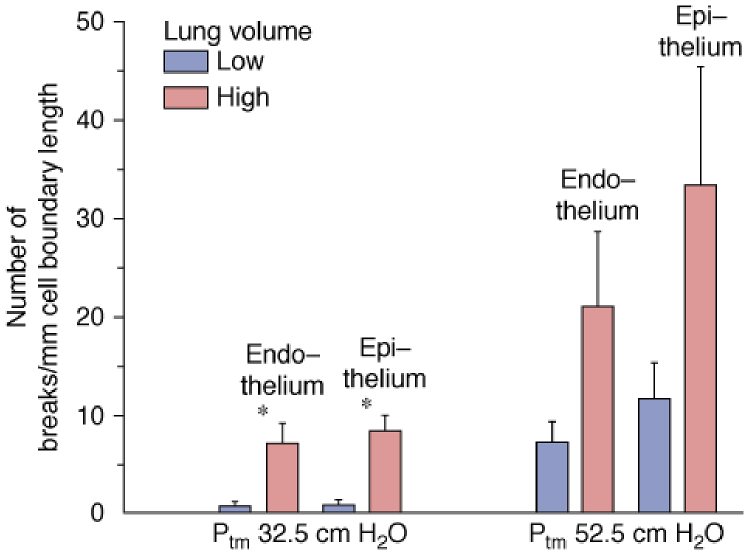
Examples of disruptions of the blood-gas barrier in in situ rabbit lungs perfused at 72.5 cm H_2O capillary transmural pressure. **A.** Transmission electron microscopy. The alveolar epithelium and capillary endothelium are disrupted, but the basement membrane is intact. **B.** Scanning electron microscopy. Circular rupture involving only the epithelial layer (*open arrow*) and complete disruption of the blood-gas barrier (*closed arrow*) showing red blood cells in the opening (*asterisk*). A large amount of proteinaceous material is present in the alveolar lumen. The *arrowhead* indicates a type II cell with an intercellular junction (*white arrow*). (Used, with permission, from West et al.[55] and Costello et al.[66])

Figure 42-12



Effect of lung inflation on capillary stress failure. In situ rabbit lungs subjected to a capillary transmural pressure of 32.5 cm H_2O had more endothelial and epithelial breaks at high lung volume (transpulmonary pressure of 20 cm H_2O) than at low lung volume (transpulmonary pressure of 5 cm H_2O). (Used, with permission, from Fu et al.[50])

Effects of Activation of Inflammation by Protracted High-Volume Ventilation

The endothelial cell disruptions that have been observed during overinflation edema in small animals may allow direct contact between polymorphonuclear cells and basement membrane (see Fig. 42-10B) and promote leukocyte activation. Proinflammatory mediators may also be released by cells undergoing necrosis. Infiltration of inflammatory cells into the interstitial and alveolar spaces is not seen before several hours of injurious ventilation. In mice subjected to high PIP ventilation, leukocyte sequestration in lungs progressively increased with time.[71] Woo and Hedley-White[72] observed that overinflation produced edema in open-chest dogs, and that leukocytes accumulated in the vasculature and macrophages in the alveoli. Further studies confirmed these results[73] and showed that high transpulmonary pressure increased the transit time of leukocytes in the lungs of rabbits.[74] Conversely, high-volume pulmonary edema was less severe in neutrophil-depleted animals. Kawano et al[75] observed that neutrophil-depleted rabbits had preserved gas exchange, little lung albumin leakage, and no hyaline membranes after 4 hours of mechanical ventilation in contrast to nondepleted animals in a saline-lavage model. Whereas it is predictable that healing of wounded tissue would involve inflammation, several authors have suggested that lung tissue stretching might result in lung damage solely through the release of inflammatory mediators and leukocyte recruitment, the so-called biotrauma hypothesis.[76]

The term *biotrauma* encompasses all molecular and cell-mediated mechanisms involved in VILI, such as the production of lung-borne cytokines.[76,77] Although the causative role of those mediators on VILI has been debated,[78] as comprehensively discussed in Chapter 42 of the second edition of this book and in "Pharmacologic Interventions" below, several investigators have tested the effect of cytokine modulators on lung dysfunction in experimental models of VILI.

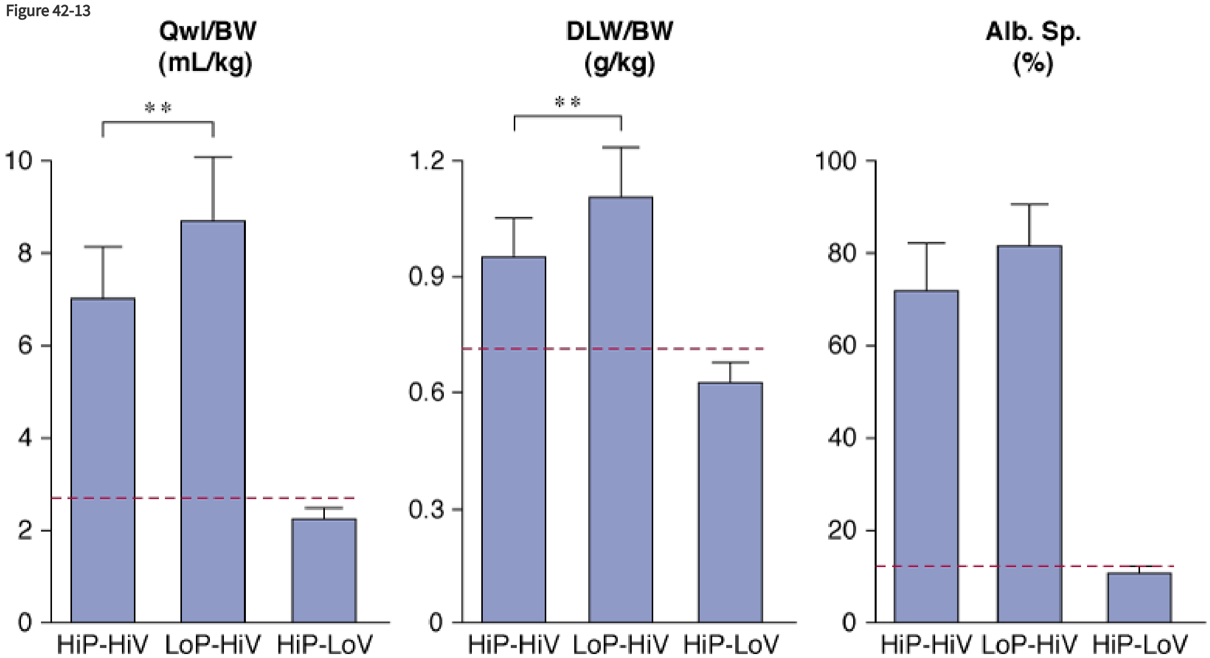
Physiologic Determinants of Ventilator-Induced Injury

Respective Roles of Increased Airway Pressure and Increased Lung Volume

High inspiratory pressures consistently result in high lung volumes in normal animals. They induce well-documented hemodynamic changes in the systemic circulation,[79] and may potentially affect blood-flow distribution in the lungs because parts of the lungs are placed under zone 1 condition over the ventilatory cycle. Another possibility is that high pressures per se produce regional distortions that may result in specific deleterious effects. Several studies have been conducted to determine the relative roles of intrathoracic pressure increases and lung distension in the genesis of pulmonary edema.

High-volume ventilation with low airway pressure was obtained by ventilating closed-chest rats with intermittent negative perithoracic pressures (using an iron lung). Ventilation with high airway pressures but low tidal volumes was obtained with thoracoabdominal strapping to limit thoracic movements. The effects of high PIP plus high tidal-volume ventilation were compared with those of intermittent negative airway pressure plus high tidal-volume ventilation and intermittently positive high PIP plus low tidal-volume ventilation.[36] Pulmonary edema of the permeability type occurred in both groups of rats that received high tidal-volume ventilation, irrespective of inspiratory pressure (i.e., positive or negative; Fig. 42-13). In both groups, electron microscopic examination disclosed the same abnormalities as described above. In striking contrast with these findings, animals ventilated with a high PIP but a normal tidal volume had no edema (Fig. 42-13) and no ultrastructural changes. These findings have been replicated in rabbits[80] and lambs.[37]

Figure 42-13



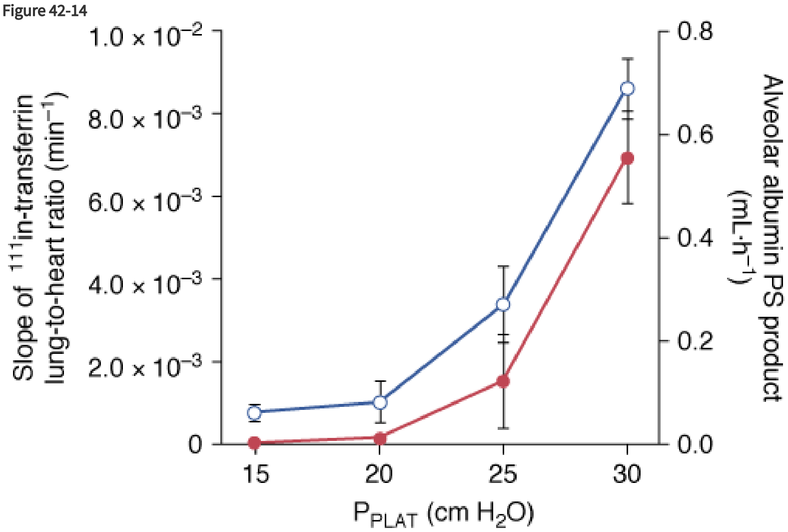
Comparison of the effects of high peak (45 cm H_2O) positive inspiratory pressure plus high tidal-volume ventilation (*HiP-HiV*) with the effects of negative inspiratory airway pressure plus high tidal-volume ventilation (iron lung ventilation = *LoP-HiV*) and of high peak (45 cm H_2O) positive inspiratory pressure plus low tidal-volume ventilation (thoracoabdominal strapping = *HiP-LoV*). *Dotted lines* represent the upper 95% confidence limit for control values. See Figure 42-6 for details on edema indexes. Permeability edema occurred in both groups receiving high tidal-volume ventilation. Animals ventilated with a high peak pressure and a normal tidal volume had no edema. ** = $p < 0.01$. (Used, with permission, from Dreyfuss et al.[36])

Roles of the Magnitude of Pressure–Volume Changes and of Duration of the Challenge

Is There a Pressure–Volume Threshold for the Occurrence of Edema and Permeability Alterations?

A threshold volume for permeability changes as the lung is overexpanded was suggested by Carlton et al.[37] These authors studied the effects of graded increases in tidal volume on lymph flow and protein concentrations in lambs. Peak inspiratory pressure (and therefore tidal volume) was increased in three successive steps (each lasting 4 hours) from baseline (16 cm H_2O to 61 cm H_2O). During the first two steps (33 and 43 cm H_2O PIP), no changes in lymph flow or protein composition were observed. In contrast, when the highest PIP was reached (corresponding to a tidal volume of 57 mL/kg), lung lymph flow increased fourfold to sixfold compared with baseline. Similarly, the lymph-to-plasma-protein-concentration ratio did not change until the highest PIP was reached, at which time it increased significantly compared with controls. The investigators concluded that microvascular alterations in response to overinflation occurred beyond a pressure threshold rather than gradually as pressure increased. They pointed out, however, that the albumin-to-globulin ratio in lymph versus plasma decreased before maximum PIP was reached, suggesting altered protein sieving and probably abnormal microvascular permeability. Indeed, Tsuno et al[39] demonstrated that ventilation of sheep with a moderately elevated PIP of 30 cm H_2O (corresponding to a tidal volume of 30 mL/kg) for more than 40 hours invariably resulted in gross pathologic alterations and an increase in wet-lung weight. Ventilating healthy sheep with tidal volumes ranging from 9 to 51 mL/kg for 54 hours, Protti et al demonstrated that ventilator-induced lung edema developed only when a strain (i.e., the ratio between tidal volume and FRC[81]) greater than 1.5 to 2.0 was applied to the lungs.[82] Interestingly, systemic inflammation and organ dysfunction also developed in pigs ventilated above this strain threshold.[82] In smaller species, such as rats, in which VILI occurs more rapidly, edema was demonstrated with a PIP as low as 30 cm H_2O, provided ventilation was sufficiently prolonged.[2] Recently, scintigraphic methods allowing for real-time imaging of two-way protein fluxes across the alveolo-capillary barrier were used to measure simultaneous changes in alveolar and microvascular permeability during lung inflation in rats.[51] The same end-inspiratory pressure threshold (between 20 and 25 cm H_2O, corresponding to tidal volumes of 13.7 ± 4.69 and 22.2 ± 2.12 mL/kg) was observed for epithelial and endothelial permeability changes (Fig. 42-14). Of note, this threshold corresponds approximately to the pressure at which a decrease is observed in the slope of the respiratory system pressure–volume curve (the so-called upper inflection point) in rats.[83]

Figure 42-14



Relationship between plateau pressure (P_{PLAT}) and ^{111}In-transferrin lung-to-heart ratio slopes (*left axis, open circles*), reflecting lung-endothelial permeability, and alveolar ^{99m}Tc-albumin permeability-surface area product (*right axis, full circles*), reflecting lung-epithelial permeability. There is an increase in the permeability to proteins in both layers of the alveolo-capillary membrane above the same P_{PLAT} threshold, between 20 and 25 cm H_2O. (Used, with permission, from de Prost et al.[51])

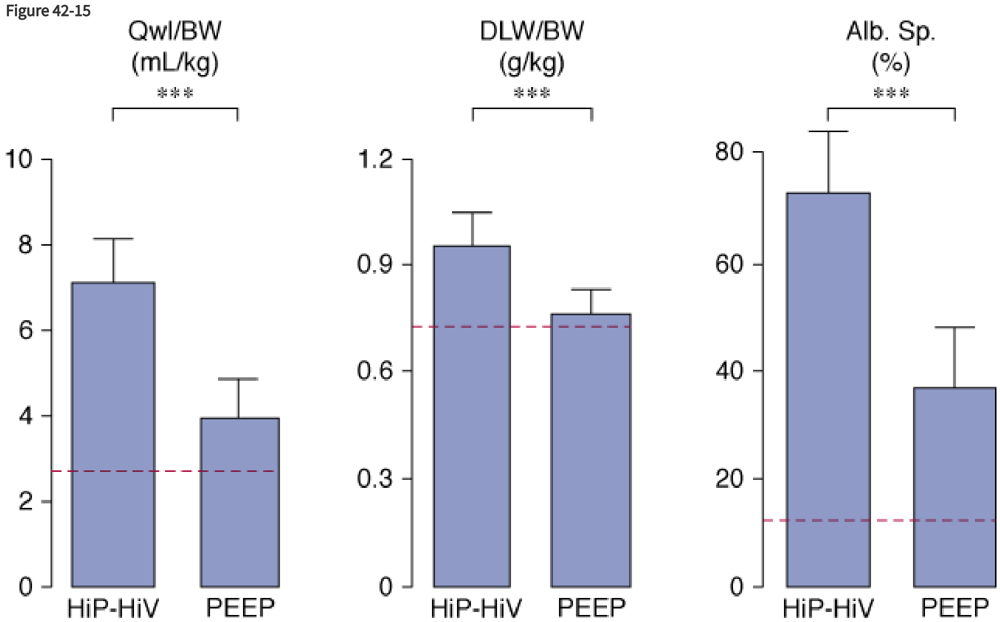
These studies emphasize that both the degree and the duration of lung overexpansion are crucial in determining the severity of pulmonary edema. Interestingly, the possibility of a potential pressure–volume threshold for additional lung injury during mechanical ventilation of patients with acute lung injury is also debated. This is discussed in "Clinical Relevance" below.

Respective Contributions of Sustained Inflation (Positive End-Expiratory Pressure) and Large Pressure–Volume Swings

For the same level of overall lung inflation, increasing FRC by PEEP results in less-severe alterations, the reasons for which may be reduction of V_T at the same respiratory rate, and hence of peak inspiratory flow, or stabilization of terminal units.

Several studies assessing the deleterious effects of high gas flow rates on lung function, a minor determinant of VILI, are discussed in detail in Chapter 42 of the second edition of this book. Webb and Tierney showed that, for a given level of teleinspiratory pressure, edema was less severe when a 10-cm H_2O PEEP was applied.[2] It was subsequently shown[36] that, although the amount of edema fluid was smaller with PEEP (Fig. 42-15), permeability alterations were similar to those observed with zero end-expiratory pressure. When rats were ventilated with PEEP, however, less edema occurred and, in particular, no alveolar flooding was observed on light microscopy.[2,36] More strikingly, whereas diffuse alveolar damage was present in the animals ventilated with zero end-expiratory pressure, no epithelial-cell lining alterations were observed on electron microscopic examination of the lungs of animals ventilated with PEEP. The only ultrastructural alterations consisted of endothelial blebbing. It is worth noting that PEEP decreases lung-capillary blood volume,[84] which, in turn, may lessen the capillary permeability alterations caused by high PIP ventilation.[85]

Figure 42-15



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

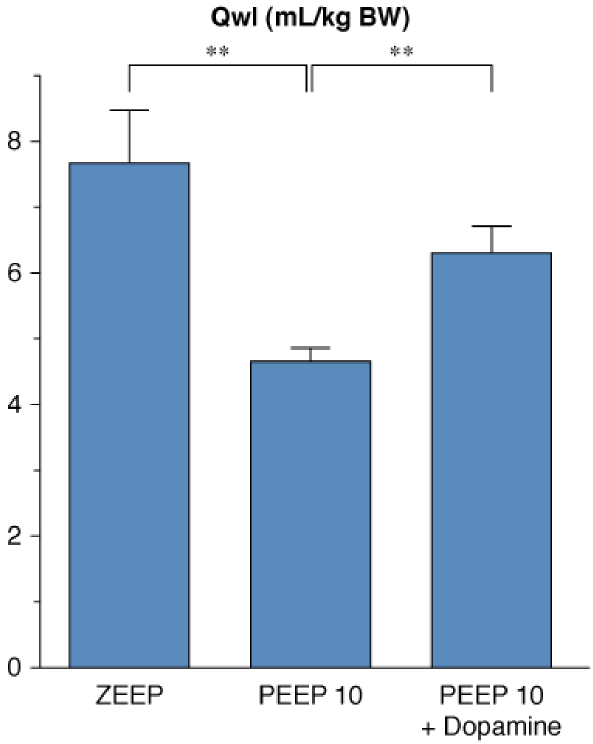
Effect of 10 cm H_2O PEEP during high peak (45 cm H_2O) positive inspiratory pressure plus high tidal-volume ventilation. *Dotted lines* represent the upper 95% confidence limits for control values. See Figure 42-6 for details on edema indexes. With PEEP, edema was less marked. ** = $p < 0.001$. (Used, with permission, from Dreyfuss et al.[36])

This preservation of the alveolar–epithelial layer has received no satisfactory explanation. It may be that PEEP eliminated repetitive opening and closing of terminal airways, thereby decreasing shear stress and the development of surface-tension-induced epithelial cell damage at this level.[86] Alternatively, avoidance of alveolar flooding by PEEP[87,88] may have protected the epithelial lining from injury through the effects of yet unidentified humoral mediators.

One effect of PEEP is to reduce tidal volume for a given end-inspiratory pressure. It is interesting to note that studies using in situ perfused canine lobes[89] found that, for equivalent perfusion-flow rates and microvascular hydrostatic pressures, the rate of hydrostatic edema formation increased with tidal volume. When an identical increase in mean airway pressure was achieved either by applying PEEP or by increasing tidal volume, edema was less marked under the former condition, suggesting that large cyclic changes in lung volume promote the development of edema. This explanation was also put forward by Corbridge et al,[90] who observed in hydrochloric acid-injured dog lungs that ventilation with a large tidal volume and a low PEEP resulted in more severe edema than did ventilation with a small tidal volume and a high PEEP. The effect of the amplitude of tidal volume on alveolar epithelium protein permeability was confirmed in rats using noninvasive scintigraphic techniques.[91] The alveolar albumin permeability-surface area product, measured from the clearance of an intratracheally instilled ^{99m}Tc-labeled albumin solution, dramatically increased, and in a dose-dependent manner, when V_T was increased from 8 to 24 and 29 mL/kg.[91]

Finally, the potential role of hemodynamic alterations during PEEP ventilation should also be considered. For instance, rats ventilated with 45 cm H_2O PIP and 10 cm H_2O PEEP had more edema when PEEP-induced hemodynamic alterations were corrected by dopamine administration (Fig. 42-16).[92] Moreover, arterial blood pressure was found to be significantly correlated with the amount of pulmonary edema under such conditions. The reason why use of PEEP during ventilation with high PIP is associated with reductions in both the amount of edema and the severity of cell damage may be a combination of hemodynamic alterations, shear stress reduction, and surfactant modifications. It should be borne in mind that this beneficial effect of PEEP during overinflation edema contrasts with the usual lack of reduction or even increase in edema reported with PEEP in most forms of experimental pulmonary edema.[93,94]

Figure 42-16



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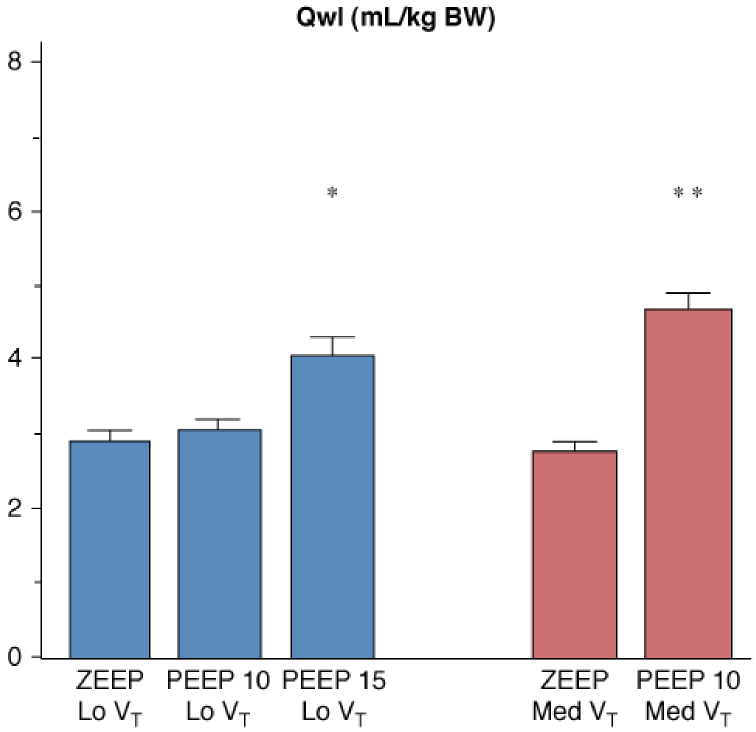
Effect of hemodynamic support with dopamine during 45 cm H_2O peak pressure ventilation with 10 cm H_2O PEEP on the amount of edema as evaluated by extravascular lung water. Compared with animals ventilated with 45 cm H_2O peak pressure and zero end-expiratory pressure (ZEEP), animals ventilated with 45 cm H_2O peak pressure ventilation with 10 cm H_2O PEEP had less edema. This reduction of edema associated with PEEP was partly abolished when dopamine was administered. ** = $p < .01$. (Adapted, with permission, from Dreyfuss and Saumon.[92])

It would be inappropriate to conclude from the data summarized above that tidal volume, which governs opening and closing of terminal units, is the sole determinant of VILI. On the contrary, the overall degree of lung distension (i.e., teleinspiratory volume) is probably the crucial factor. Hence, rats ventilated with a tidal volume within the physiologic range at two levels of PEEP (10 and 15 cm H_2O) developed pulmonary edema only at the higher level of PEEP (Fig. 42-17).[92] Similarly, doubling tidal volume had no effect in animals ventilated with zero end-expiratory pressure, but resulted in pulmonary edema when 10 cm H_2O PEEP was used (Fig. 42-17).[92] Thus, the safety of small tidal volumes depends on whether or not FRC is increased, and raising end-inspiratory volume by increasing FRC may cause lung injury independently of tidal volume.[92]

Figure 42-17



Effect of increasing PEEP from 0 to 15 cm H_2O during ventilation with two levels of tidal volume (V_T, 7 mL/kg of body weight [BW] = *Lo* V_T; 14 mL/kg BW = *Med* V_T). When PEEP was increased, pulmonary edema (as evaluated by extravascular lung water increases) occurred. The level of PEEP required to produce edema varied with tidal volume: 15 cm H_2O PEEP during ventilation with a low tidal volume versus 10 cm H_2O PEEP during ventilation with a moderately increased V_T. * = $p < .05$; ** = $p < .01$ versus zero end-expiratory pressure (*ZEEP*) and the same V_T. (Used, with permission, from Dreyfuss and Saumon.[92])

It is now clear that VILI occurs whenever a certain degree of lung overinflation is reached, by whatever means. For a given level of teleinspiratory pressure (and volume), adjunction of PEEP seems to slow the development or diminish the severity of alterations[95] but does not prevent the occurrence of permeability pulmonary edema.[36,92]

Effects of Ventilation on Previously Injured Lungs

This chapter does not consider the problems associated with lung prematurity, which is discussed in Chapter 23.

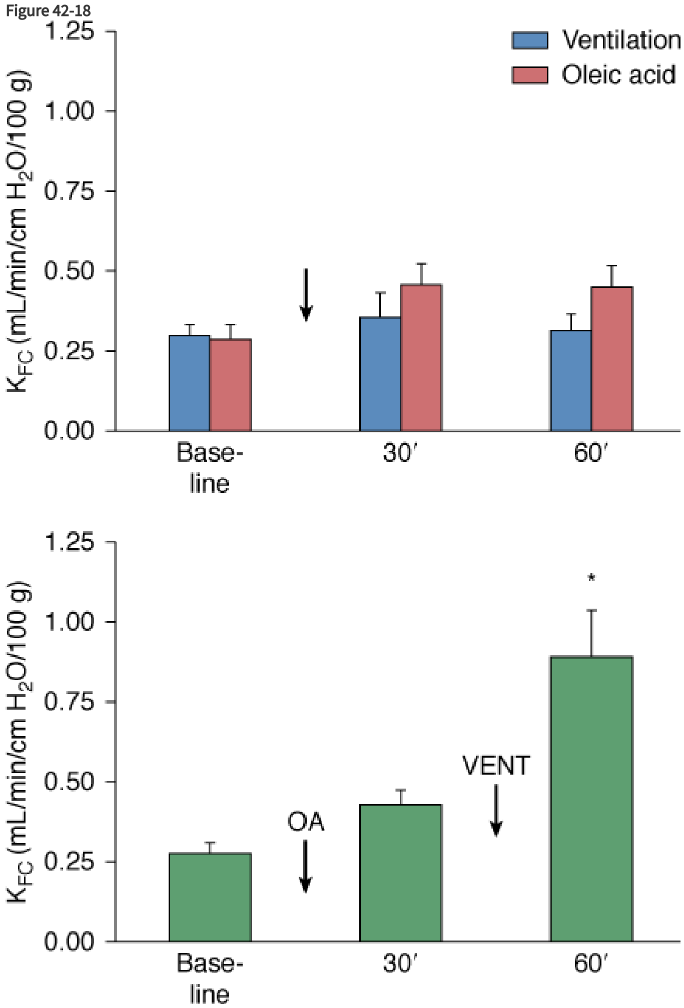
Combination of High-Volume–High-Pressure Ventilation and Previous Injury

Role of Lung Mechanical Properties on Susceptibility to VILI

The aforementioned studies were conducted on animals with healthy lungs. It is conceivable, however, that diseased lungs may be more susceptible than healthy lungs to the deleterious effects of mechanical ventilation. Diseased lungs with a patchy distribution of lesions may be subjected to considerably greater regional stress than homogeneously inflated lungs. As stressed by Mead et al, the pressure tending to expand an atelectatic region surrounded by a fully expanded lung is approximately 140 cm H_2O at a transpulmonary pressure of 30 cm H_2O.[96]

Hernandez et al[97] showed that whereas low doses of oleic acid or mechanical ventilation with PIP 25 cm H_2O did not affect filtration coefficient and wet-to-dry ratio, the combination of the two did (Fig. 42-18). The same group also reported that the filtration coefficient increase observed during high PIP (30 to 45 cm H_2O) ventilation of isolated perfused rabbit lungs was more marked following inactivation of surfactant by dioctyl succinate instillation.[98] Moreover, whereas light microscopic examination showed only mild abnormalities (minimal hemorrhage and vascular congestion) in the animals subjected to ventilation only or surfactant inactivation only, the combination of both injuries caused severe damage (extensive hemorrhage, pulmonary edema, and hyaline membranes). Thus, ventilator-induced lung edema seems to develop at lower airway pressures in lungs with preexisting injury. Lung injury is usually inhomogeneous. The more compliant zones will thus receive the bulk of ventilation, which may favor their overinflation, a localized "baby lung effect." Analysis of the pressure–volume curve may help understand how preexisting lung injury may interact with ventilator-induced injury. Before examining this interaction, it is important to understand how lung mechanical properties are modified by lung injury.

Figure 42-18

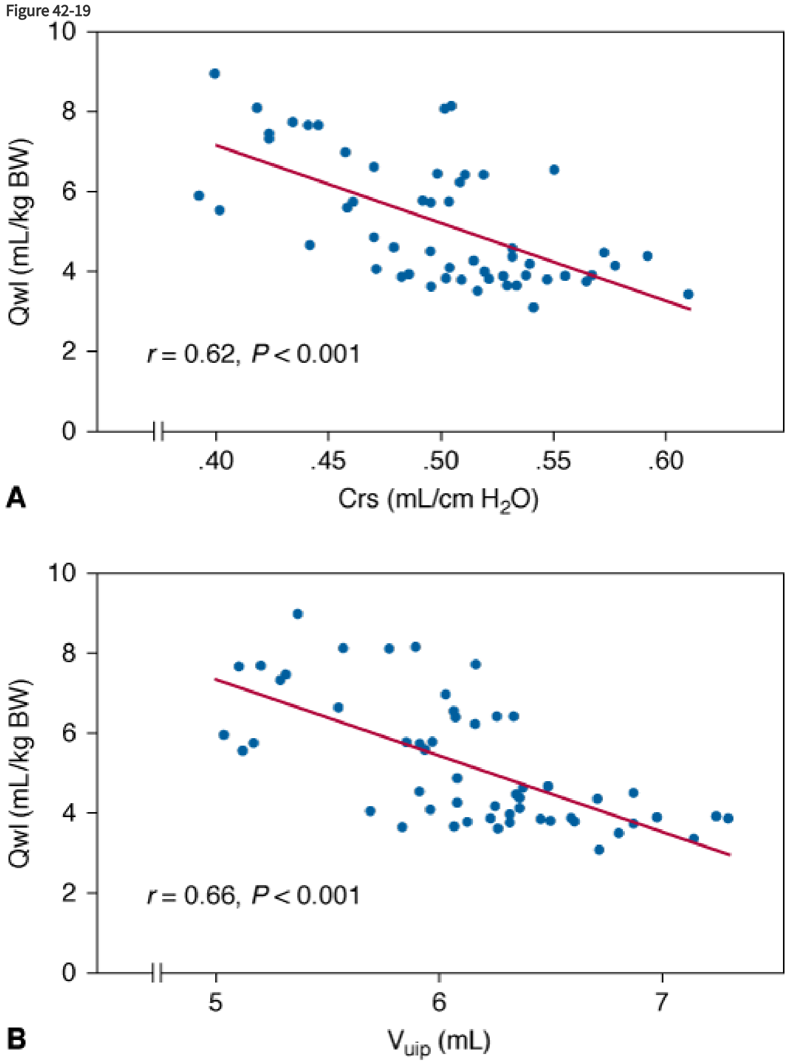


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Effect of single and combined insults on the filtration coefficient of isolated rabbit lungs. A low dose of oleic acid (*OA*) or a moderately high peak pressure (24 cm H_2O) alone failed to induce K_{FC} (capillary filtration coefficient) changes, whereas the combination of both insults was responsible for a significant increase in K_{FC}. (Used, with permission, from Hernandez et al.[97])

The upper inflection point (UIP) often seen on the inspiratory pressure–volume curve of the respiratory system in patients with ARDS has been interpreted as reflecting the beginning of overinflation,[99,100] or the end of recruitment.[101,102] Whether ventilation, however, that results in pressure and/or volume excursions above the UIP is deleterious is unsettled. Better understanding of the UIP significance is required before it can be used to set tidal volume in patients. An experimental study examined the hypothesis that pulmonary edema development alters the pressure–volume curve of respiratory system mainly because of distal airway obstruction,[83] and this reduction in ventilatable lung volume (the baby lung effect) may not only decrease compliance[103,104] but also affects UIP position. When the distal airways of rats were obstructed by instilling a viscous liquid, the changes in the shape of the pressure–volume curve (gradual decrease in compliance, the volume at which UIP was seen and progressive increase in end-inspiratory pressure) were similar to the changes seen during the development of pulmonary edema. The higher the compliance and volume of the UIP, the lesser was the edema observed after high PIP ventilation (Fig. 42-19). Taken together, these results suggest that the position of the UIP is a marker of the amount of ventilatable lung volume, and it is both influenced by, and predictive of, the development of edema during mechanical ventilation.

Figure 42-19

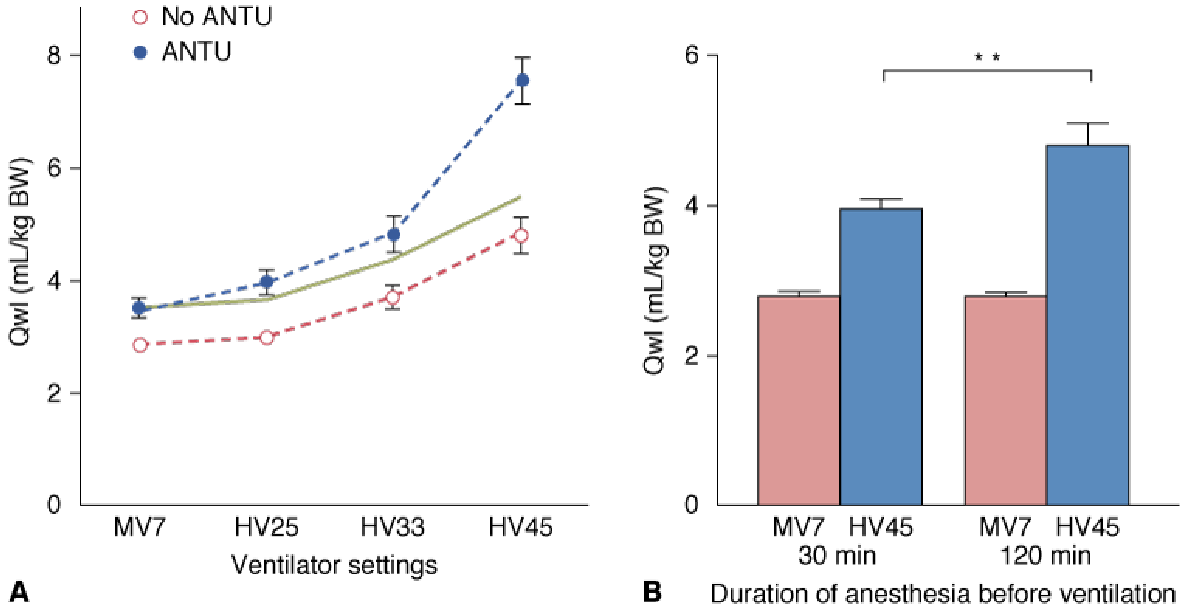


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

A. Correlation between the amount of edema (extravascular lung water, *Qwl*) produced by high-volume ventilation and the respiratory system compliance (*Crs*) measured before high-volume ventilation. **B.** Correlation between *Qwl* and the volume at the upper inflection point on the pressure–volume curve (V_{uip}) measured before high-volume ventilation. (Used, with permission, from Martin-Lefèvre et al.[83])

The effect of high PIP ventilation on injured lungs in intact animals was investigated by comparing different degrees of lung distension in rats whose lungs had been injured by α-naphthylthiourea (ANTU).[105] ANTU infusion alone caused moderate interstitial pulmonary edema of the permeability type. Mechanical ventilation resulted in a permeability edema, the severity of which depended on the tidal-volume amplitude. It was possible to calculate the extent that mechanical ventilation theoretically injures lungs diseased by ANTU by adding the separate effect of mechanical ventilation alone or ANTU alone on edema severity (Fig. 42-20). The lungs of the animals injured by ANTU ventilated at high PIP (45 mL/kg body weight [BW]) had more severe permeability edema than predicted (Fig. 42-20), indicating synergy between the two insults. Even minor alterations, such as those produced by spontaneous ventilation during prolonged anesthesia (which degrades surfactant activity and promotes focal atelectasis), were sufficient to synergistically increase the harmful effects of high-volume ventilation.[105] The extent to which lung mechanical properties deteriorate before ventilation is a key factor in this synergy. The amount of pulmonary edema produced by high-volume ventilation in animals given ANTU, or that had undergone prolonged anesthesia, was inversely proportional to the respiratory system compliance measured at the very beginning of mechanical ventilation.[83,105] The same conclusions were reached about the volume of UIP.[83]

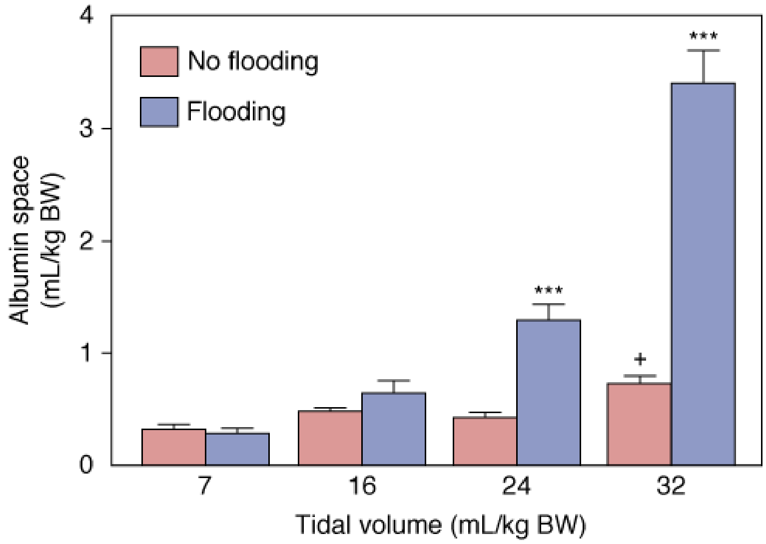
Figure 42-20



Interaction between previous lung alterations and mechanical ventilation on pulmonary edema. **A.** Effect of previous toxic lung injury. Extravascular lung water (*Qwl*) after mechanical ventilation in normal rats (*open circles*) and in rats with mild lung injury produced by α-naphthylthiourea (*ANTU*) (*closed circles*). Tidal volume (V_T) varied from 7 to 45 mL/kg BW. The *solid line* represents the Qwl value expected for the aggravating effect of ANTU on edema caused by ventilation, assuming additivity. ANTU did not potentiate the effect of ventilation with V_T up to 33 mL/kg BW. In contrast, V_T 45 mL/kg BW produced an increase in edema that greatly exceeded additivity, indicating synergy between the two insults. **B.** Effect of lung functional alteration by prolonged anesthesia. Intact rats were anesthetized and breathed spontaneously for 30 to 120 minutes before ventilation with V_T 7 mL/kg BW (*pink open bars*) or 45 mL/kg BW (*blue shaded bars*). Qwl of animals ventilated with a high V_T was significantly higher than in animals ventilated with a normal V_T . Qwl was not affected by the duration of anesthesia in animals ventilated with a normal V_T . In contrast, 120 minutes of anesthesia before high V_T ventilation resulted in a larger increase in Qwl than did 30 minutes of anesthesia. ** = $p < 0.01$. (Used, with permission, from Dreyfuss et al.[105])

The reason for this synergy requires clarification. Local alveolar flooding in animals subjected to the most harmful ventilation protocol was the most striking difference from animals ventilated with lower, less harmful, tidal volumes.[105] It is conceivable that edema foam in airways reduced the number of alveoli that received the tidal volume, exposing them to overinflation and rendering them more susceptible to injury, further reducing the aerated lung volume. The result is a positive feedback loop. To explore this possibility, alveolar flooding was produced by instilling saline into the trachea of rats that were immediately ventilated with tidal volumes of up to 33 mL/kg.[106] Flooding with saline did not significantly affect microvascular permeability when tidal volume was low. As tidal volume was increased, capillary permeability alterations were larger in flooded than in intact animals, reflecting further impairment of their endothelial barrier (Fig. 42-21). There was also a correlation between end-inspiratory airway pressure and capillary permeability alterations in flooded animals ventilated with a high tidal volume. Thus, the less compliant and recruitable the lung was after saline flooding, the more severe were the changes in permeability caused by lung distension.

Figure 42-21



Effect of increasing tidal volume (V_T) during mechanical ventilation for 10 minutes on lung capillary permeability (i.e., extravascular albumin distribution space in lungs) of rats with intact lungs (*pink open bars*) or with alveolar flooding (*purple closed bars*) produced by saline instillation. There was a moderate increase in albumin space in intact rats at the larger V_T. Lung flooding did not produce significant increases of albumin space when V_T was normal or moderately increased. Albumin space was significantly increased with a V_T of 24 and 32 mL/kg BW. The increase in albumin space greatly exceeded additivity, indicating a positive interaction between the two insults. *** = $p < 0.001$ as compared with intact animals. + = $p < 0.05$, as compared to animals of the "no flooding" group ventilated with V_T7 mL/kg. (Used, with permission, from Dreyfuss et al.[106])

These studies support the conclusion that the risk of overinflation is more important in diseased than in healthy lungs. Strategies to prevent VILI should oppose the synergy between ventilation and previous lung injury.

Does Ventilation Without Overinflation Aggravate Previous Lung Injury?

Effect of Positive End-Expiratory Pressure

It has been suggested that mechanical ventilation may worsen lung injury because of increased shear stress in distal airspaces secondary to their repeated closing, either because of obstruction by liquid menisci or collapse and reopening. Coker et al[98] reported that the capillary filtration coefficient of isolated perfused rabbit lungs was not modified by either ventilation at 15 cm H_2O PIP (which results in much larger tidal volumes in isolated lungs than in intact animals) or surfactant inactivation, but doubled the coefficient when the two insults were combined. Several studies have investigated the effects of conventional ventilation on acutely injured lungs by using sufficiently high PEEP levels to keep the small airways open throughout the entire ventilatory cycle, thereby avoiding repeated closure and reopening.

Sykes et al[107,108] studied the effect of ventilation in rabbits with surfactant-depleted lungs. The animals were ventilated with a PIP ranging from 15 mm Hg at the beginning of the experiment to 25 mm Hg at the end (5 hours later) because of the fall in lung compliance (tidal volume was not stated); PEEP adjusted to position the FRC either above or below the lower inflection point (LIP). Mortality rate was not influenced by PEEP level, although arterial oxygen tension (Pa_{O_2}) was better-preserved in the high-PEEP group.[107,108] There was less hyaline membrane formation in animals ventilated with a high PEEP than in those ventilated with low PEEP. This lessening of pathologic alterations was observed even with ventilator settings that achieved identical mean airway pressures in the low and high PEEP groups.[108] Comparable results were reported in isolated, nonperfused, lavaged rabbit lungs ventilated with a low tidal volume and with a PEEP level set either below or above the LIP.[109] The same findings could not be replicated in hydrochloric acid-injured rabbit lungs using similar ventilator settings.[110] Thus, it is conceivable that the protective effect of PEEP, when set above the LIP of the pressure–volume curve, is observed only in the very special setting of surfactant deficiency and not during severe alveolar edema, because lung instability and airspace collapse is observed only during the former.

Using in vivo videomicroscopy, Nieman et al[111–115] directly observed and quantified the dynamic changes in alveolar size throughout the ventilatory cycle during tidal ventilation, in normal lungs and in lungs in which surfactant had been deactivated by generic. Injuries occurred independently of the presence of neutrophils.[111] In normal lungs, alveoli never collapsed. These findings agree with the previous findings of Bachofen and Wilson, who showed that alveoli do not change volume appreciably during ventilation.[116] Collapse and reopening and increase in the alveolar size at end-inspiration were observed in surfactant-deactivated lungs. In a subsequent study, the same authors documented the effect of increasing end-expiratory pressure.[115] The application of PEEP to a surfactant-deactivated lung reversed the observed increase in alveolar size, returning it to control levels.

The reality of the repetitive opening and closure of terminal units and the significance of the LIP on the pressure–volume curve have been challenged by Martynowicz et al,[117] who studied the regional expansion of oleic acid-injured lungs using a parenchymal marker technique. The gravitational distribution of volume at FRC was not affected by oleic acid injury, and the injury was not associated with decreased parenchymal volume of dependent regions. In addition, temporal inhomogeneity of regional tidal expansion did not increase with oleic acid injury. These findings do not support the hypothesis that a superimposed gravitational pressure gradient during VILI produces compression atelectasis of dependent lung, which in turn produces shear injury from cyclic recruitment and collapse.[117] The authors propose that the occurrence of a LIP on the pressure–volume curve represents the transition from the liquid-filled state to the air-filled state, when the lung is initially filled with a liquid with constant surface tension properties.[118] Thus, abrupt expansion of distal lung units is unlikely to occur at the LIP.

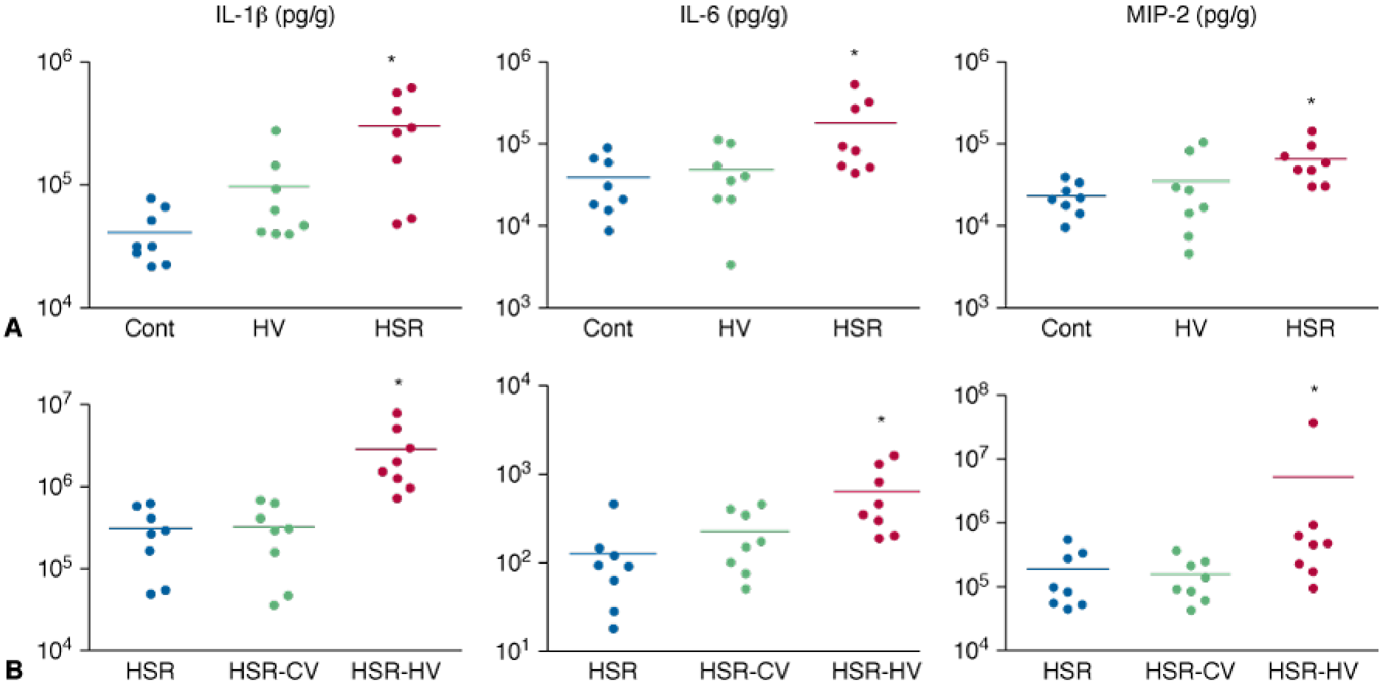
The protective effect of PEEP, however, is not restricted to the particular setting of surfactant depletion.[2,3,95] Distal airspace damage during tidal ventilation may occur because of repeated airway opening and closing secondary to the movement of foam with increased surface tension[86] or the rupture of liquid menisci. PEEP may prevent diffuse lung damage during prolonged ventilation by stabilizing these units.[119] The beneficial effect of PEEP, however, is variable in different models of lung injury.[120] The difficulty in proving whether or not PEEP exerts a protective effect on VILI may be explained by taking into account the distinction between atelectatic and fluid-filled distal airways when interpreting LIP. In the case of diffuse filling of distal airways with liquid, VILI may be caused by overdistension of already aerated zones rather than by shear stress secondary to opening of collapsed zones.[121] With this scenario, it would be difficult to expect any reduction of injury by PEEP.

This uncertainty about the actual occurrence of injury at low lung volumes contrasts with the unambiguous demonstration of high-volume (overdistension) injury. This uncertainty has a clinical counterpart that is discussed in “Clinical Relevance” below.

Effect on Remote Organs and Normal Lung Regions

As further discussed in “Pharmacologic Interventions” below, neutrophilic infiltration and increased levels of proinflammatory cytokines have been found in the lungs of animals that were subjected to long-lasting injurious ventilation.[73,76] These observations led to the hypothesis that mechanical ventilation might promote or aggravate a systemic inflammatory state.[122] In addition to increasing the amount of cytokines in the lung, it has been suspected that overinflation may promote the release of cytokines[123,124] or bacteria[125–127] into the blood, thus suggesting that mechanical ventilation plays a causative role in multiorgan dysfunction.[122,128] Higher levels of proinflammatory cytokines (tumor necrosis factor [TNF]-α, interleukin [IL]-1β and macrophage inflammatory protein [MIP]-2) were found in lung homogenates of rats ventilated for 2 hours with high PIP when they were previously subjected to mesenteric ischemia–reperfusion[129] or hemorrhagic shock and resuscitation[130] (Fig. 42-22). This suggests the validity of the “two-hit” hypothesis, that a preexisting inflammatory state augments cytokine release during VILI.

Figure 42-22

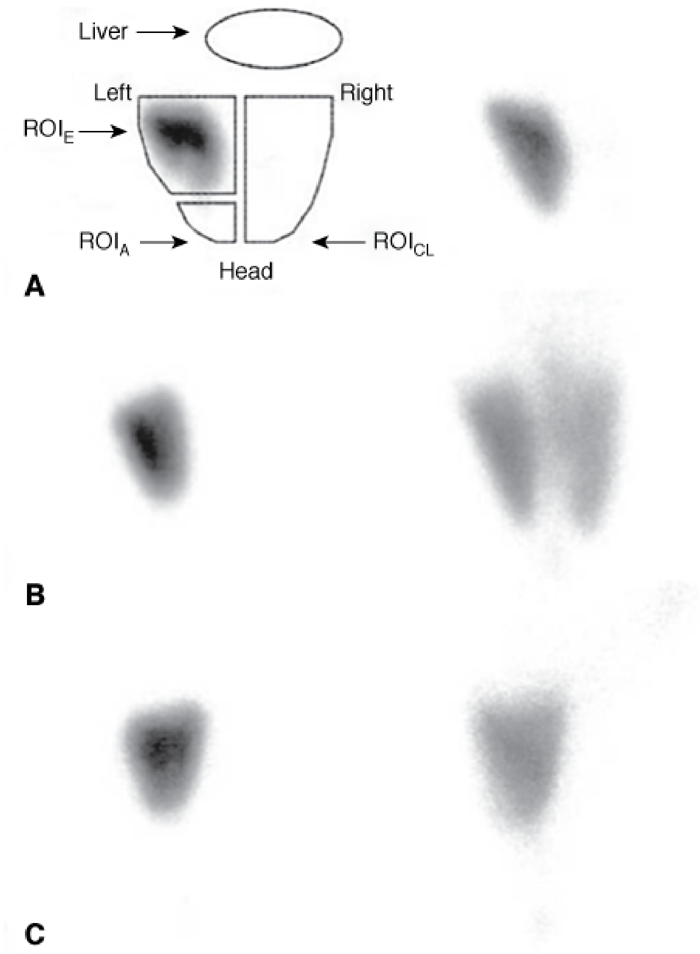


A. Comparison of lung cytokine levels in nonventilated rats (controls [*Cont*]) and in rats subjected to an injurious mechanical ventilation strategy (30 mL/kg tidal volume and zero end-expiratory pressure) alone (*HV*) or to hemorrhagic shock-reperfusion injury alone (*HSR*). Compared with controls, lung cytokine concentrations were higher after HSR but not after HV. **B.** Comparison of lung cytokine levels in rats subjected to HSR alone, HSR combined with conventional ventilation (*HSR-CV*) and HSR followed by HV (*HSR-HV*). Injurious ventilation (HV) after HSR significantly increased mediator release above the levels observed after HSR alone or in combination with conventional ventilation. Results are expressed in pg/g. * = $p < 0.05$ as compared to controls (*A*) or to HSR (*B*). The same observations were made in bronchoalveolar lavage fluid and in plasma. (Adapted, with permission, from Bouadma et al.[130])

A few experimental studies have evaluated the consequences of injurious ventilation on peripheral organs. Choi et al[131] found that rats ventilated with high PIP for 2 hours had increased endothelial nitric oxide synthase expression in lung and kidney tissue and increased microvascular permeability in both organs. Nitric oxide was probably involved in these abnormalities, because *N*-nitro-l-arginine methyl ester administration attenuated the microvascular leak of lung and kidney. An increase in gut permeability was also observed during a similarly long-lasting ventilation, albeit with larger tidal volume (30 instead of 20 mL/kg), which was attenuated by anti–TNF-α antibody administration.[132] Injurious ventilation of rabbits with hydrochloric acid-induced lung injury resulted in increased rates of epithelial cell apoptosis in the kidney and small intestine villi.[133]

There is also growing evidence that ventilator settings may localize or disperse proteinaceous lung edema or bacteria. In a model of unilateral *Pseudomonas aeruginosa* pneumonia in rats, Schortgen et al showed that high-volume ventilation with no PEEP promoted contralateral bacterial seeding.[127] Interestingly, ventilation at the same end-inspiratory pressure but with high PEEP, and thus a lower V_T, prevented contralateral lung dissemination. To better understand these observations, the potential effect of adverse ventilator patterns on the dispersal of localized radiolabeled alveolar edema to the opposite lung was studied.[91] A ^{99m}Tc-labeled albumin solution was instilled into the distal airways and it produced a zone of alveolar flooding that remained localized during conventional ventilation. High end-inspiratory pressure ventilation dispersed alveolar liquid in the lungs. This dispersion was prevented by PEEP even when V_T was the same and thus end-inspiratory pressure even higher (Fig. 42-23). Interestingly, contralateral liquid dispersion began almost immediately after high-volume ventilation was started, suggesting that this dispersion might be the consequence of a convective movement induced by the ventilation. High-inspiratory and high-expiratory flows favored fluid transport back and forth toward the alveoli and the airways. In the heterogeneous lung, fluid transfer may be propelled toward regions of normal compliance. PEEP may have prevented dispersion by avoiding lung collapse and stabilizing edema fluid in the distal airways. The clinical relevance of the movement of such noxious biofluids in the airways has been discussed elsewhere.[134]

Figure 42-23



Scintigraphy images integrated over 15 minutes after instillation (t0 to t15; *left panels*) and the last 15 minutes of the experiment (t195 to t210; *right panels*). Regions of interest (*ROI*s) were drawn around initial focus of edema (ROI_E), the apex of the same lung (ROI_A), the contralateral lung (ROI_{CL}), and over the thorax. At baseline (*left panels*), all animals were ventilated with tidal volume 8 mL/kg and PEEP 2 cm H_2O and exhibited focalized localization of the tracer in the left lung. When the same ventilator settings were maintained during the experiment (**A**), the tracer remained remarkably confined in the initial zone; there was no contralateral and only slight homolateral dissemination. High-volume ventilation (pressure plateau [P_{PLAT}] = 30 cm H_2O) with no PEEP (**B**) induced strong homolateral and contralateral dispersion of the tracer and systemic leakage as attested by the decrease in overall activity. High-volume ventilation with 6 cm H_2O PEEP (*C*) induced systemic, but not contralateral, dissemination of the tracer. (Used, with permission, from de Prost et al.[91])

Protection from Ventilation-Induced Lung Injury

Strategies that Aim at Improving Lung Mechanics

Use of Pressure–Volume Curves

Taking the presence and value of a LIP on the pressure–volume curve into account when setting the level of PEEP may lessen VILI in some,[107–109] but not all, instances.[110] The concept on lung protection against VILI with the setting of a PEEP above the LIP, however, relies on a putative stabilizing effect of PEEP that would abolish the tendency of distal airways to collapse at end-expiration and reopen during the next inspiration. As discussed above, serious criticisms have been raised against this opening-and-closing theory,[121] casting doubt on the usefulness of measuring LIP in order to prevent VILI.

The overall degree of lung distension resulting from the settings of both PEEP and tidal volume is a fundamental determinant of VILI. Although the exact physiologic significance of UIP is debated, it may indicate lung overstretching. Ventilation that takes place above UIP was found to be markedly deleterious (see Fig. 42-19).[83] Thus, determination of the volume at which the UIP is observed may help reduce the risk of VILI, because it indicates the maximum stretch that the lung can sustain without noticeable damage.[83] Analysis of the airway pressure–time curve during mechanical ventilation with constant flow has also proven useful in determining the stress applied to lungs. Indeed, an upward concavity of the pressure–time curve suggests that compliance decreases as tidal volume is delivered, with resulting high-volume stress.[135] Such a ventilation pattern was associated with histologic lung injury in an isolated, nonperfused, lavage model of acute lung injury[135] and with overdistension, as attested by computed tomography scan analysis, of the lungs of intact animals with saline lavage-induced lung injury.[136]

Surfactant

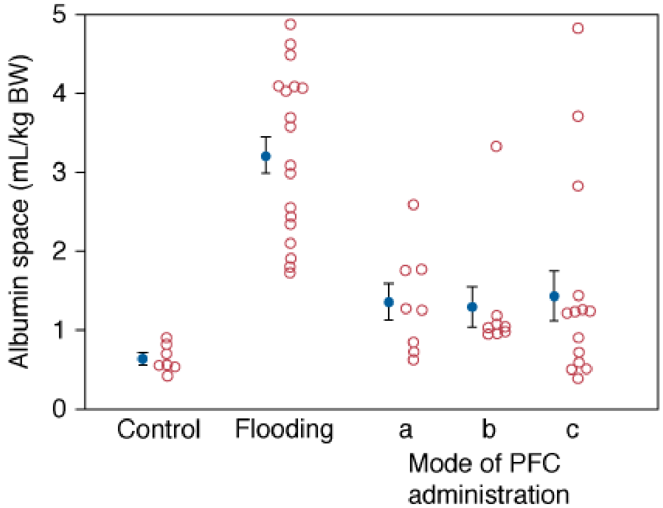
Numerous studies have been conducted on the effect of surfactant administration on gas exchange in experimental models of lung injury. In contrast, few experimental studies have specifically addressed the effect of surfactant administration during VILI. Verbrugge et al studied the effect of surfactant administration on lung function and alveolar permeability during 45 cm H_2O PIP ventilation in rats.[137] Alveolar permeability was significantly reduced in animals receiving 200 mg/kg of surfactant.[137] Exogenous surfactant prevented high-volume (20 mL/kg) lung injury in isolated rat lungs, but it did not affect the release of inflammatory cytokines by these lungs during ventilation.[138] In rats receiving lipopolysaccharide by tracheal instillation, however, surfactant pretreatment reduced decompartmentalization of TNF-α from the lungs to the systemic circulation during injurious mechanical ventilation.[139]

Perfluorocarbons

Partial liquid ventilation[140] with perfluorocarbons was developed during the last decade as an alternative to conventional gas ventilation for the treatment of acute respiratory failure. Several investigators using different models of acute respiratory failure (surfactant depletion, oleic acid, hydrochloric acid, prematurity) have reported improvement in gas exchange, lung mechanics, and lung histology.[141] Although some investigators have shown a dose-dependent improvement in gas exchange with perflubron (LiquiVent) in a rabbit model of surfactant depletion,[142] others have highlighted the risk of barotrauma (namely pneumothorax) with the use of large volumes of perfluorocarbon combined with high PEEP or increased tidal volume.[143] Until recently, the effect of partial liquid ventilation on VILI has received little attention. Mechanical nonuniformity of diseased lungs may predispose lungs to VILI by overinflation of the more compliant (ventilatable), aerated zones.[105] Perfluorocarbon may reduce this nonuniformity by suppressing air–liquid interfaces and allowing reopening of collapsed or liquid-filled areas.

The effect of perfluorocarbon instillation on hyperinflation-induced lung injury during alveolar flooding was investigated in rats.[106] Saline was instilled into the trachea to mimic alveolar edema and reduce aerated lung volume. Alveolar flooding significantly aggravated VILI, as attested by an increase in capillary permeability alterations. Tracheal instillation of a low dose (3.3 mL/kg) of perflubron in these flooded lungs considerably reduced VILI and decreased permeability alterations (Fig. 42-24). Whereas instillation of saline alone raised LIP pressure to values as high as 25 cm H_2O, and produced a significant increase in the end-inspiratory pressure, administration of perflubron significantly reduced LIP pressure and normalized end-inspiratory pressure.

Figure 42-24

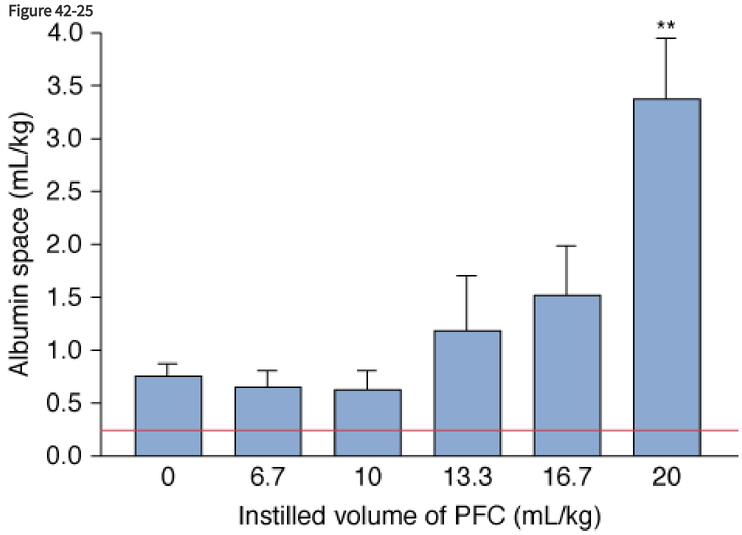


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Effect of perflubron (LiquiVent) instillation on permeability pulmonary edema as assessed by the extravascular albumin distribution space in the lungs of rats ventilated with a tidal volume of 33 mL/kg BW. Flooding significantly increased albumin space ($p < 0.001$). Perflubron given as a bolus (*a*), by slow infusion before flooding (*b*), or as a bolus dose after flooding (*c*) resulted in a significant decrease in albumin space, the values of which remained higher than in controls ($p < 0.05$). *Closed circles with error bar* indicate means ± standard error of mean (SEM). *PFC,* perfluorocarbon. (Used, with permission, from Dreyfuss et al.[106])

To further investigate the effect of partial liquid ventilation on VILI with respect to the dosage of perfluorocarbon, intact animals ventilated with a high tidal volume (33 mL/kg BW) received increasing doses of perflubron (from 6 to 20 mL/kg).[144] Hyperinflation-induced pulmonary edema tended to decrease with doses of perflubron lower than 10 mL/kg as compared with animals not given perflubron. By contrast, ventilator-induced pulmonary edema was aggravated in animals given 13 and 16 mL/kg, and even more in animals given 20 mL/kg perflubron (Fig. 42-25). Large doses worsened volutrauma because they increased FRC, thus increasing end-inspiratory volume for the same tidal volume, and because they favored gas trapping in the distal lung, as demonstrated by computed tomography imaging.[144] Consistently, in rats with preinjured lungs (after ANTU administration) and subjected to ventilation with a high tidal volume (33 mL/kg BW), low and moderate doses of perflubron, but not larger doses (approximately 20 mL/kg), improved respiratory mechanics and ventilator-induced permeability alterations (Fig. 42-26).[145] These observations suggest that monitoring of end-inspiratory pressure might help detect the risk of volutrauma during partial liquid ventilation.

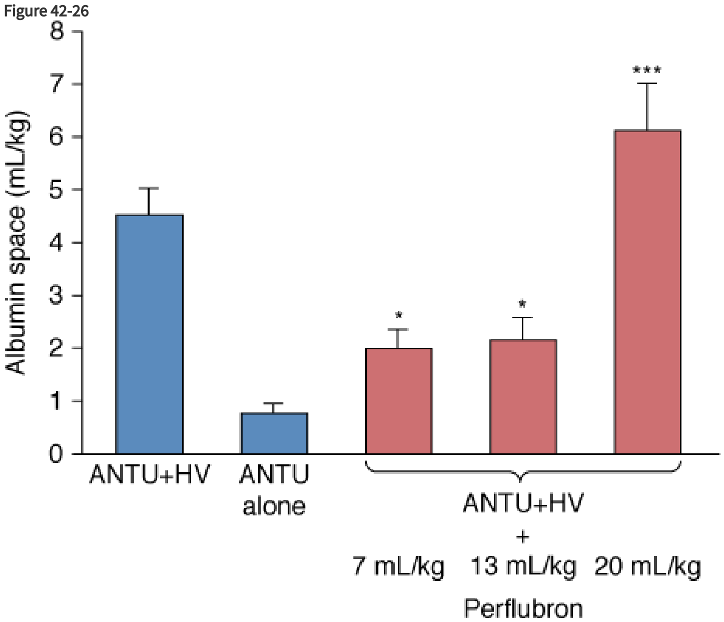
Figure 42-25



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Dose–response effect of perflubron instillation on microvascular permeability assessed by the distribution space of albumin in the lung. Rats were ventilated with a tidal volume of 33 mL/kg BW. The albumin space value of control rats ventilated with a tidal volume of 7 mL/kg BW is represented by the *horizontal line.* Albumin space was three times higher after high-volume ventilation, in the absence of perflubron instillation. Lung injury was not aggravated by the instillation of 6.7 and 10 mL/kg perflubron, tended to increase with doses of 13.3 and 16.7 mL/kg, and increased significantly with 20 mL/kg perflubron. ** = $p < 0.01$; *PFC,* perfluorocarbon. (Used, with permission, from Ricard et al.[144])

Figure 42-26



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Dose–response effect of perflubron instillation on microvascular permeability assessed by the distribution space of albumin in lungs injured either by α-naphthylthiourea (*ANTU*) alone or combined with high tidal-volume ventilation (tidal volume 33 mL/kg BW; *ANTU+HV*). ANTU alone produced a mild permeability defect, which was considerably increased when combined with HV. This dramatic change in microvascular permeability was significantly reduced by doses of 7 and 13 mL/kg perflubron (* = $p < 0.05$). In contrast, animals that received 20 mL/kg perflubron had a further significant increase in microvascular permeability, as compared with animals that received ANTU+HV without perflubron instillation (*** = $p < 0.001$). (Used, with permission, from Ricard et al.[145])

Prone Position

Prone position lessened the deleterious effect of high PIP ventilation with PEEP in dogs with previous oleic acid lung injury;[146] similar benefit was shown in rabbits[147] and rats.[148] Prone position resulted in a more homogenous distribution of lung injury,[149] an observation that was not confirmed in other studies in rabbits or rats. These differences are probably a result of difference in lung size, with a larger lung increasing the influence of gravity. Indeed, computed tomography scans in rats did not disclose a gradient of lung inflation at end-expiration as in humans.[148] In rats, the beneficial effect of prone position was ascribed to a more homogenous distribution of tidal volume and thus of strain, because of the downward displacement of the diaphragm.[148] In a rat model of paraquat-induced acute lung injury, the prone position was consistently associated with a more homogeneous lung expression of type III procollagen than observed in the supine position.[150]

Noisy Ventilation

Spontaneously breathing subjects show an intrinsic variability in tidal volume and respiratory frequency during normal ventilation.[151] It has been theorized that the respiratory system may work as a stochastic resonance system where the variability of input parameters leads to an increase in the surface area for gas exchange in the lungs, thus improving lung function.[152] In a model of endotoxin-induced acute lung injury in guinea pigs, variability of tidal volumes of some 10% to 60% above and below the mean combined with a respiratory frequency that kept minute ventilation constant significantly decreased lung elastance and improved oxygenation as compared to conventional ventilation.[153] After oleic acid-induced lung injury, pigs ventilated with a comparable strategy exhibited lower concentrations of IL-8 in tracheal aspirates than did pigs ventilated with a conventional strategy.[154] Recently, Spieth et al[155] studied the effects of variability of tidal volume (within

40% of mean) in a surfactant-depletion model of acute lung injury in pigs. Animals were ventilated with two different strategies for setting PEEP: the ARDS Network protocol and an open-lung approach where PEEP was set according to the minimal elastance of the respiratory system. Adding noisy ventilation to these settings not only improved oxygenation, but also decreased histologic damage.[155] Depending on the strategy with which it was combined, noisy ventilation likely improved VILI by distinct mechanisms. When combined with the ARDS Network protocol, noisy ventilation stabilized distal lung units and likely reduced their repetitive opening and closing. When combined with an open-lung approach, noisy ventilation generated lower peak airway pressures, thereby limiting the overdistension of nondependent lung regions. Further experimental studies are required to determine whether similar results would be obtained in less recruitable models of acute lung injury and after longer periods of mechanical ventilation.

Pharmacologic Interventions

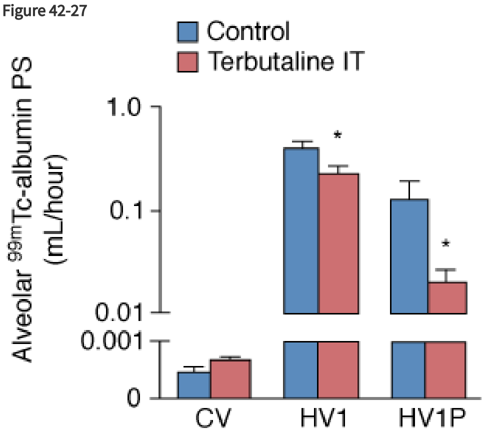
The following observations show that a tremendous number of cell-signaling pathways are involved in the pathophysiology of VILI. Although it is probably illusory to believe that a single pharmacologic intervention might be beneficial in patients, the description of those pathways illustrates the complexity of the cellular mechanisms involved in VILI.[78]

Modulation of Microvascular Permeability

Blocking stretch-activated cation channels[52] and inhibiting phosphotyrosine kinase,[54] phosphodiesterase,[54] and calcium-dependent phospholipase A_2[71] reduced the pulmonary capillary permeability alterations caused by high PIP ventilation. Reducing myosin light-chain phosphorylation with adrenomedullin, an endogenous peptide belonging to the calcitonin gene-related peptide family, decreased lung vascular permeability and the accumulation of neutrophils in the lungs of mice subjected to high PIP ventilation.[156] These inhibitions suggest that stretch-induced calcium entry in lung cells and intracellular calcium signaling play an important role in the early response to high PIP ventilation. Indeed, the phosphorylation of myosin light chains by Ca^{2+}/calmodulin-dependent myosin light chain kinase leads to endothelial cell contraction, a key mechanism of VILI-induced endothelial barrier dysfunction.[157] Intravenous delivery of a myosin light-chain kinase inhibitor in a mouse model of VILI reduced lung microvascular permeability and neutrophilic infiltration.[158]

β-adrenergic agonist infusion reduced accumulation of plasma protein in the lungs of patient who survived an episode of ARDS,[159] suggesting that it might have lessened their microvascular permeability alterations. In a model of high PIP ventilation in rats, terbutaline administered intratracheally lessened both the increase in lung epithelial and endothelial permeability caused by VILI, whereas terbutaline administered intraperitoneally lessened only lung endothelial permeability (Fig. 42-27).[160] These results help us understand the disappointing results of a trial that compared the intravenous administration of β-adrenergic agonists versus placebo in patients with ARDS.[161] In that study, gas exchanges and lung injury score remained unchanged despite a decrease in extravascular lung water, as measured by thermodilution. The findings suggest that airway delivery might be more efficient in preventing the increase in alveolocapillary barrier permeability during mechanical ventilation.

Figure 42-27



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

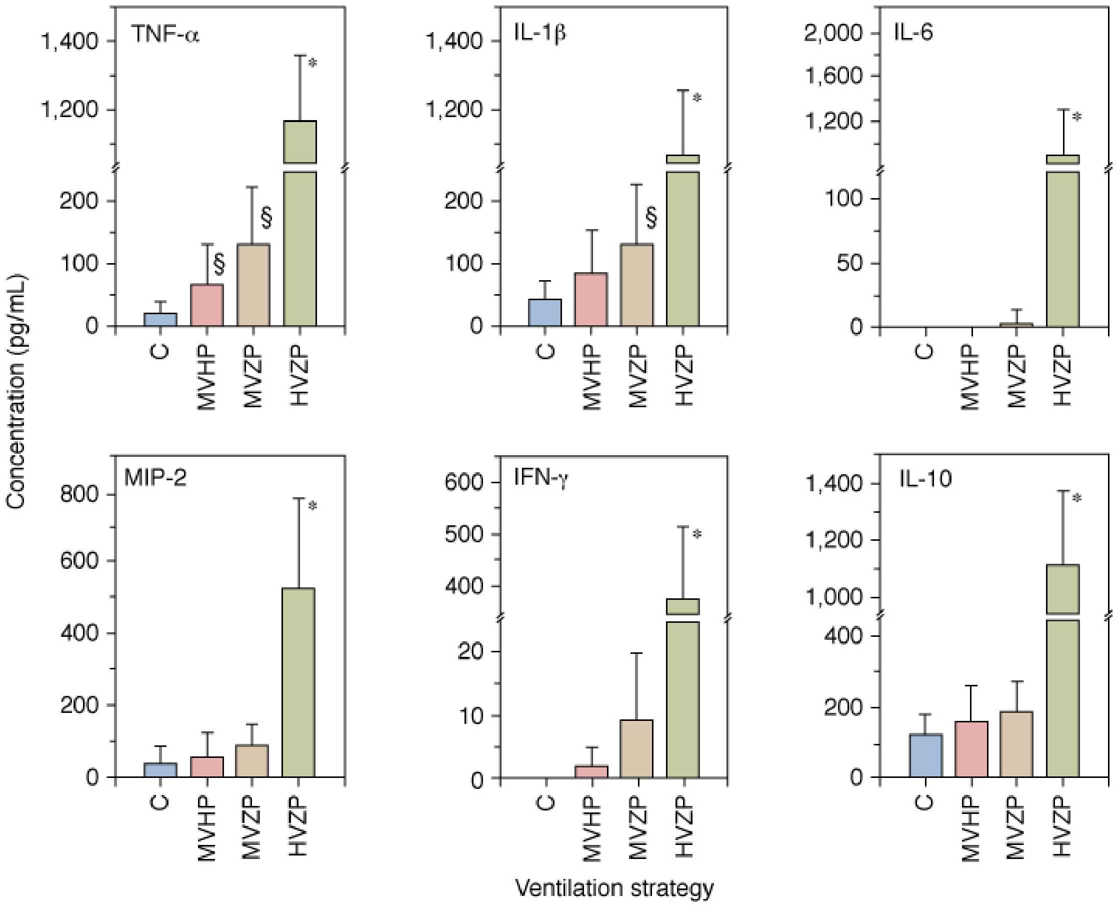
Alveolar ^{99m}Tc-albumin permeability surface-area product (*PS*), reflecting lung-epithelial permeability. PS was dramatically increased during high-volume ventilation using 30 cm H_2O PIP (*HV1*), as compared to conventional ventilation (*CV*). This effect was present even when positive end-expiratory pressure was applied (*HV1P*) and was decreased when terbutaline was administered intratracheally (*IT*). * = $p < 0.05$ when compared with the same ventilation modality. (Adapted, with permission, from de Prost et al.[160])

Inhibition of nitric oxide synthase[131,162] lessened ventilator-induced capillary permeability abnormalities. Inhibition of the endothelial isoform mitigated VILI via a decrease in the production of superoxide.[163] Recently, phosphoinositide 3-kinase-γ of resident lung cells was consistently shown to mediate alveolar edema induced by high-stress ventilation in part through an increase in nitric oxide production by endothelial cells.[164] Surprisingly, transgenic mice that overexpressed endothelial nitric oxide synthase exhibited less lung damage, lung water content, and neutrophil infiltration as compared to wild-type mice, suggesting that this isoform might also be protective from VILI in different conditions.[165] These discrepant results suggest that the mechanisms of VILI are equivocal and that more work is needed to determine the involvement of the nitric oxide pathway despite the fact that the ultrastructural abnormalities observed after high-volume ventilation[3] resemble those seen during increased endothelial nitric oxide production.[166]

Modulation of Inflammation

Tremblay et al[76] examined the effects of different ventilator strategies on the level of several cytokines in bronchoalveolar lavage fluid of ex vivo, unperfused rat lungs ventilated with different end-expiratory pressures and tidal volumes. High tidal volume ventilation (40 mL/kg BW) with zero end-expiratory pressure resulted in the release of considerable amounts of TNF-α, IL-1β, IL-6, and MIP-2, a potent neutrophil chemoattractant and the rodent functional homolog of human IL-8 (Fig. 42-28). These results have not, however, been replicated by other groups using either the same ex vivo lung model.[78,167] In vivo studies of intact animals showed that high-volume mechanical ventilation, which produces a very severe pulmonary edema, does not induce lung release of TNF-α in rats.[167,168] Similarly, TNF-α release was not found in mice ventilated with a high PIP in vivo, although it was found in the perfusate of isolated lungs from the same strain.[30] Studies on TNF-α messenger RNA (mRNA) also yielded conflicting results. Takata et al[169] showed large increases in TNF-α mRNA in the intraalveolar cells of surfactant-depleted rabbits after 1 hour of conventional mechanical ventilation with peak inspiratory and end-expiratory pressures of 28 and 5 cm H_2O. Conversely, Imanaka et al found no increase in lung tissue TNF-α mRNA of rats ventilated with 45 cm H_2O PIP.[170] Administration of an anti–TNF-α antibody via the trachea, but not via the systemic circulation, lessened inflammation in lungs of mice subjected to high PIP ventilation.[132,171,172] Mice with knockout of the TNF receptor had less lung inflammation, but no less release of CXC chemokines, in bronchoalveolar lavage fluid.[30]

Figure 42-28



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Effect of different ventilator strategies on cytokine concentrations in lung lavage of isolated unperfused rat lungs. Four ventilator settings were used: controls (*C* = normal tidal volume), moderate tidal volume + high PEEP (*MVHP*), moderate tidal volume + zero PEEP (*MVZP*), high tidal volume + zero PEEP (*HVZP*) resulting in the same end-inspiratory distension as MVHP. Major increases in cytokine concentrations were observed with HVZP. * = $p < 0.05$ vs all other groups; § = $p < 0.05$ vs controls. (Used, with permission, from Tremblay et al.[76])

The only mediator that was consistently released by lungs subjected to high-volume ventilation[30,76,123,167,173] or in vitro in stretched lung cells, such as human alveolar macrophages[174] or cell lines such as A549 epithelial cells,[175] was the chemokine IL-8 or its rodent equivalent MIP-2. The release of MIP-2 may explain the recruitment of neutrophils that occurs during subacute VILI.[73,176,177] Leukocyte sequestration during injurious ventilation strategies seems to be the result of polymorphonuclear stiffening and mediated by L-selectin–dependent but CD18 (β_2-integrin)-independent mechanisms.[178] Belperio et al[173] showed that inhibition of the chemokine interaction with the CXCR2 receptor attenuates neutrophil sequestration and lung injury after 6 hours of high-volume ventilation in mice. Whether neutrophil recruitment alone is sufficient to produce VILI or merely accompanies mechanical injury continues to be debated.[78] Inhibition of MIP-2 activity by specific antibodies[179] of the MIP-2 receptor[173] reduced neutrophil infiltration and lung injury caused by high PIP ventilation.

Leukocyte activation, however, is an unlikely explanation for the major vascular leakage observed during high-inflation edema because the speed with which permeability and ultrastructural alterations develop[4] is not consistent with such a mechanism. Besides, the potent proinflammatory mediators, TNF-α and MIP-2, are unlikely to be involved in the acute lung injury consequent to high-volume ventilation[30] for two reasons. First, there was no correlation between the flux of albumin in alveolar spaces (as reflected by its concentration in bronchoalveolar lavage fluid) and MIP-2 (or IL-6) concentration in this fluid. Second, the authors did not detect TNF-α in bronchoalveolar fluid in any intact mouse model that developed pulmonary edema after lung overinflation and the absence of TNF-α receptors did not preclude the development of acute VILI.[30] These observations confirm that two different types of VILI exist: acute VILI that follows marked overexpansion, and subacute VILI in which inflammatory phenomena prevail. Recently, the use of genomic-intensive approaches allowed for the identification of VILI susceptibility candidate genes[180] and revealed that pre–B-cell colony-enhancing factor is a direct neutrophil chemotaxin and potentially a key pharmacologic target in VILI.[181]

The concept of biotrauma (i.e., the release of proinflammatory cytokines and mediators by lung parenchymal cells, alveolar macrophages, and neutrophils) led to the publication of a tremendous number of studies that tested the effect of modulating the imbalance between proinflammatory and antiinflammatory mediators in the lung. For instance, ventilation with high tidal volume of isolated mouse lungs for 1 hour resulted in nuclear factor-κB (NF-κB) activation, which was inhibited by dexamethasone.[182] Pretreatment of rats with NF-κB antibodies resulted in less permeability pulmonary edema, pathologic changes, and IL-1 expression in an isolated rat lung model of VILI.[183] Administration of an IL-1 receptor antagonist reduced lung albumin, elastase, and neutrophil counts in surfactant-depleted rabbits ventilated with hyperoxia and a high tidal volume[184] and decreased pulmonary edema, airspace neutrophils, and expression of nitric oxide synthase 2 and intercellular adhesion molecule 1 mRNA in rats ventilated with 30 mL/kg tidal volume.[185] In a rat model of VILI, Hoegl et al showed that the prophylactic inhalation of IL-22 before initiation of 45 cm H_2O PIP ventilation protected the lung against lung edema and improved survival.[186] This beneficial effect might be related to an increase in STAT3/SOCS3 expression in alveolar epithelial cells. In a rat model of VILI, prophylactic IL-10 inhalation mitigated the increase in MIP-2 and IL-1β in bronchoalveolar fluid and plasma, and increased animal survival time.[187]

Finally, other kinds of antiinflammatory interventions, such as the stimulation of the endogenous cholinergic pathway,[188] the administration of adenosine A_{2A} receptors agonists,[189] the inhalation of carbon monoxide,[190] and the inhibition of the cyclooxygenase[191] or lipooxygenase[192] showed beneficial effects in animal VILI models.

Modulation of Hormonal and Metabolic Pathways

The renin–angiotensin system is involved in lung inflammation during VILI, as attested by increased angiotensin II levels and angiotensinogen mRNA levels in bronchoalveolar lavage fluid of rats subjected to 4 hours of high PIP ventilation.[193] Angiotensin-converting enzyme II, which inactivates angiotensin II and is thus a negative regulator of the renin–angiotensin system, protects mice from acute lung injury related to acid aspiration or sepsis.[194] Moreover, pretreatment of animals with captopril attenuated lung injury and inflammation.[193,195] These protective effects might be related to a reduction in the angiotensin II-induced production of plasminogen activator inhibitor-1.[196]

Pretreatment with atorvastatin decreased alveolar capillary permeability and improved hemodynamics in an isolated rabbit lung model of VILI.[197] Muller et al confirmed these findings in an in vivo model of VILI in mice, showing that treatment with simvastatin limited lung endothelial injury and hyperpermeability, decreased lung neutrophilic infiltration, and improved oxygenation.[198] It remains unknown, however, whether these effects are related to the lipid-lowering properties of statins or to antiinflammatory, antioxidative, or nitric oxide synthesis regulatory properties.[196]

Lung-soluble mediators are pathogenic in VILI.[199] Indeed, isolated perfused mouse lungs ventilated with low tidal volumes but perfused with a perfusate derived from lungs previously ventilated with high tidal volumes develop a similar decrease in compliance and an increase in lung water.[199] Several lipid-derived mediators have been identified as candidate mediators in VILI.[200] For instance, inhibition of cytosolic phospholipase A_2 mitigates the increase in capillary permeability induced by high PIP ventilation.[201] Finally, ceramides and other sphingolipids are essential constituents of plasma membranes and have been recognized as potential pharmacologic targets in the treatment of acute lung injury.[202] Whether this also applies to prevention or treatment of VILI is unknown.

Reversible induction of a suspended animation-like state (i.e., a hypometabolic state characterized by decreased heart rate, body temperature, and exhaled amounts of CO_2) obtained by the intravenous infusion of a hydrogen sulfide donor, reduced pulmonary inflammation and improved oxygenation in rodent models of VILI.[203,204]

Hypercapnia

Ventilator strategies consisting of a low tidal volume to lower the risk of VILI have a positive impact on survival in ARDS.[1] It has been proposed that the hypercapnic acidosis that accompanies a tidal volume reduction may reduce VILI.[205]

Extreme hypoventilation (with partial pressure of arterial carbon dioxide [Pa_{CO_2}] between 150 and 250 mm Hg) lessened lung injury in surfactant-depleted rabbits, suggesting a therapeutic role for high Pa_{CO_2}.[206] Studies on isolated rabbit lungs[205] found that hypercapnia lessened the acute (within 30 minutes) increase in endothelial permeability produced by high PIP ventilation. Other studies in rabbits ventilated for 4 hours showed that hypercapnic acidosis lessened neutrophil infiltration and lung injury when tidal volume was high,[207] and also reduced the increases in alveolar–arterial oxygen gradient and airway pressure at a clinically relevant tidal volume.[208] Two other in vivo studies, however, were unable to demonstrate a reduction by hypercapnia of the acute capillary permeability alterations caused by high tidal-volume ventilation in a rabbit model of surfactant depletion[209] or in intact rats.[210] The latter investigation concluded that hypercapnia probably has a greater influence on late inflammation-dependent phenomena than on acute stress failure.

Studies in isolated rat lungs, however, have shown that hypercapnia lessened lung weight gain and the decrease in compliance during high tidal-volume ventilation,[211] but the number of subpleural damaged cells, evaluated by confocal microscopy with a membrane impermeant fluorescent tracer, did not differ between normocapnic and hypercapnic lungs. Furthermore, compared to normocapnia, hypercapnia significantly reduced the probability of wound repair in A549 cell cultures.[211] These observations suggest that there is no strong correspondence between the occurrence of cell lesions and the physiologic response of the lung to stretch. They also suggest that the benefit from hypercapnia, if any, may be offset by alterations in the cellular reparation process.[49] This hypothesis was further confirmed by Caples et al in an isolated-perfused rat-lung model of VILI.[212] The buffering of hypercapnic acidosis decreased the rate of cell membrane wounding when lungs were exposed to an injurious ventilation strategy, which suggested the importance of pH-dependent mechanisms.

Clinical Relevance

It is clear that microvascular lung injury and pulmonary edema during mechanical ventilation are not the consequences of “barotrauma,” but rather of “volutrauma.”[36] The main determinant of volutrauma seems to be end-inspiratory volume (the overall lung distension) rather than tidal volume or FRC (which is dependent on the level of PEEP).[92]

It is fascinating to see how the experimental concept of VILI was rapidly translated into a preoccupation of clinicians, as exemplified by the term *ventilator-associated lung injury.*[213] Although the validity of this concept could not be directly demonstrated in patients, it formed the rationale beneath all lung-protective strategies that aimed to reduce the risk of lung overinflation during ventilation of patients with ARDS. That the lungs in ARDS exhibit significant heterogeneity is well documented,[103,214] and ARDS is a condition that results in uneven distribution of ventilation. Gattinoni et al[103] demonstrated that ARDS lungs include healthy tissue, recruitable tissue, and diseased tissue unresponsive to pressure changes. Healthy units represent as little as 20% to 30% of total units.[103] These units can be viewed as “baby lung,” another term for the “shrunken lung” described by Gibson and Pride in patients with lung fibrosis.[104] Thus, during conventional ventilator treatment in ARDS, the bulk of ventilation may be directed to healthy units, resulting in regional overdistension, especially when high PIP is required.

Ventilator strategies aimed at reducing the risk of overinflation, such as extracorporeal membrane oxygenation,[215] extracorporeal CO_2 removal,[216] and high-frequency oscillation[217] were proposed (see Chapters 19, 20, 21). These techniques are merely experimental (at least in adults) and require special devices and highly trained physicians and nurses. In addition, the lack of demonstrable benefit with extracorporeal CO_2 removal,[218] and the scarcity of data on high-frequency oscillation in adults,[219] preclude their acceptance in many centers. Emerging data, however, have generated renewed interest in these techniques.[220,221] It is beyond the scope of this chapter to discuss this issue extensively. Nevertheless, the complexity of these techniques is in striking contrast with the extraordinary simple lung-protective strategies that accompany permissive hypercapnia.[222] Although caution should be exercised when extrapolating experimental data to clinical situations as complex as infant respiratory distress syndrome or ARDS, the clinical relevance of available experimental data received resounding support with the publication of the ARDS Network study.[1] This study showed that simply reducing tidal volume from 12 mL/kg to 6 mL/kg resulted in a 22% reduction in mortality. The effectiveness of this strategy was attributed to a reduction of lung stress, as attested by markedly decreased airway plateau pressure.[223] The demonstration, however, of a technique to prevent overdistension is difficult. Airway pressure monitoring is easy to perform but cannot completely eliminate the risk of VILI. For instance, application of a continuous distending pressure during high-frequency oscillation or extracorporeal CO_2 removal may be associated with a gradual increase in lung volume. If the increase in volume is the result of recruitment of previously closed lung units, it is likely to be beneficial, at least in terms of gas exchange, and will probably not cause additional lung damage. On the contrary, failure to recruit closed alveoli may result in overdistension of open alveoli.[224]

Because of the study protocol, the same reduction in tidal volume was employed in all patients allocated to the low tidal volume group in the ARDS Network trial.[1] It has, however, repeatedly been shown that the pressure and volume considered safe for some patients with ARDS may cause lung overdistension in others.[99,100,103,225] Conversely, arbitrary settings may result in an unnecessary reduction in tidal volume.[226] As discussed above, information from the inspiratory pressure–volume curve of the respiratory system may be used to tailor ventilator settings, especially when the significance of its particular characteristics (LIP, UIP) are better understood.[83]

Another question that cannot be answered at present is whether or not a threshold tidal volume exists below which experimental VILI or ventilator-associated lung injury in patients does not occur in previously injured lungs. If such a threshold exists, it is unnecessary to drastically reduce tidal volume in all patients as was done in the ARDS Network study,[1] especially in view of possible adverse effects that occur with too great a reduction in tidal volume.[226] In such conditions, staying within the so-called safe limits of plateau pressure[213,223] will be sufficient. In contrast, the absence of such a safe threshold was suggested in the ARDS Network study:[1] Mortality was lower in patients with reduced tidal volume independently of static compliance of the respiratory system at baseline. This observation suggests that low tidal volume was advantageous regardless of lung compliance.[1] Notwithstanding, the reduction of mortality by simply reducing end-inspiratory lung stress lends credence to the possibility that end-inspiratory lung volume is the main determinant of volutrauma (as contended at the beginning of this section). Indeed, it was recently suggested that higher tidal volume may be associated with the onset of acute lung injury[227] or ARDS[228] in patients ventilated for other causes of respiratory failure.

In contrast with the experimental and clinical demonstration of the importance of reducing lung stretch by reducing tidal volume, the clinical relevance of the concept of low lung-volume injury (discussed above) is less obvious. This is a crucial issue as it governs the level of PEEP that should be applied to reduce ventilator-associated lung injury. It is beyond the scope of this chapter to discuss in detail the clinical studies that evaluated this issue. It is worth noting, however, that three randomized, controlled trials did not show any difference in survival of patients with ARDS ventilated with higher or lower levels of PEEP.[229–231] Despite the conclusion of a meta-analysis of these three studies[232] that higher PEEP levels were associated with better survival in the patient with the most severe diseases, the best level of PEEP to apply when ventilating patients with ARDS is still unknown,[233,234] given the fact that ventilator protocols were markedly different between the three studies, with considerable heterogeneity in the values for plateau pressure in the high PEEP groups.

A recent study measuring lung metabolic activity (a surrogate of lung inflammation), using positron emission tomography with ^{18}F-fluorodeoxyglucose (^{18}F-FDG) in patients with ARDS showed that lung regions undergoing cyclic recruitment–derecruitment did not exhibit higher ^{18}F-FDG uptake than did regions that remained collapsed throughout the respiratory cycle.[235] Instead, ^{18}F-FDG uptake in normally aerated regions correlated with plateau pressure and regional volume expansion. Although these results should be interpreted cautiously because of the small number of patients involved and because pulmonary infection, a major confounding factor for the interpretation of lung inflammation, was often the cause of ARDS, they suggest that "volutrauma" rather than "atelectrauma" is a more significant contributor to lung inflammation in ventilated patients with ARDS.[235]

Improving lung mechanical properties so as to decrease the risk of ventilator-associated lung injury is another option that may combine with protective lung strategies. Two treatments were tested during clinical trials: surfactant administration and partial liquid ventilation. The failure of both strategies could have been predicted on physiologic and experimental grounds.

Administration of surfactant by aerosol[236] or tracheal instillation[237] was not associated with any survival advantage in the treatment of adult patients with ARDS. These treatments failed to improve oxygenation in one study[236] and achieved only mild improvement in another study[237] and were not associated with changes in lung mechanics.[237] It has been suggested that the treatments were not effective in preventing damage caused by lung overstretching or oxygen toxicity, either because the mode of administration was not optimal or because the particular type of surfactant was not physiologically effective in diseased lungs. In contrast, administration of a natural surfactant containing high levels of surfactant-specific protein B in a pediatric population with acute lung injury (premature infants were not included) was associated with increased survival.[238] Moreover, the oxygenation index (which takes into account both lung mechanical properties and oxygenation) was markedly improved.

The recent failure of a multicenter clinical trial of partial liquid ventilation with perflubron (a perfluorocarbon) during mechanical ventilation of adult patients with acute lung injury illustrates the clinical relevance of VILI.[239] The amount of administered perflubron was high and was associated with a high level of PEEP (>13 cm H_2O). Consequently, inspiratory pressure was higher in the patients who received perflubron than in controls. Not only was mortality higher (although not significantly) in patients receiving liquid ventilation, but the incidence of macroscopic barotrauma was also particularly high, more than double that in the controls (17% vs. 6%). All these results were predictable from careful analysis of experimental studies, which showed increased incidence of barotrauma[143] and worsening of VILI[106,144,145] when both the amount of instilled perfluorocarbon and the pressures delivered by the respirator were high.

As already discussed, the literature on possible cytokine involvement during VILI contains many contradictions and inconsistencies.[78] Indeed, it is not surprising that the clinical correlate of such experimental findings is very vague. Ranieri et al[77] reported significant decreases in bronchoalveolar fluid and plasma concentrations of many inflammatory and antiinflammatory mediators in patients ventilated with a so-called lung-protective strategy (low tidal volume plus high PEEP). They suggested that nonprotective ventilation (high tidal volume plus low PEEP) was responsible for the inflammatory state. In contrast, Stuber et al[240] reported the release of mainly antiinflammatory mediators in the bronchoalveolar fluid and plasma of patients ventilated with a nonprotective modality.[241] Finally, very modest although significant decreases of plasma inflammatory cytokines were reported in patients of the low tidal-volume arm of ARDS Network study:[1] compared with patients of the high tidal-volume arm.[242] The decrease, however, was trivial and very unlikely to have any clinical consequence. Thus, developing possible antiinflammatory therapy to prevent ventilator-associated lung injury based on these findings is speculative at best.

Conclusion

Considerable clinical progress has accrued from the extensive physiologic experimental research on VILI since its initial description by Webb and Tierney in 1974.[2] There is now widespread consensus on the need to ease the stress on diseased lungs during mechanical ventilation. The extent, however, to which tidal volume should be decreased, the level of PEEP that should be used, and how to identify lung overdistension in an individual patient remain largely unknown. A tremendous number of experimental studies aimed at exploring the pathophysiology of VILI and at identifying pharmacological targets for its prevention and treatment have been performed. None of those pharmacologic interventions, however, has proven beneficial in patients. The multiplicity of the pathways involved in VILI suggests that a single pharmacologic intervention is unlikely to impact on the outcome of patients.

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